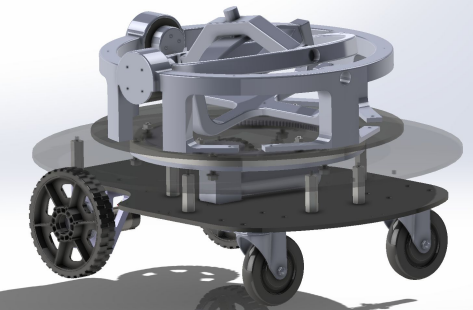




HAMR 3.0: Off-Road Holonomy

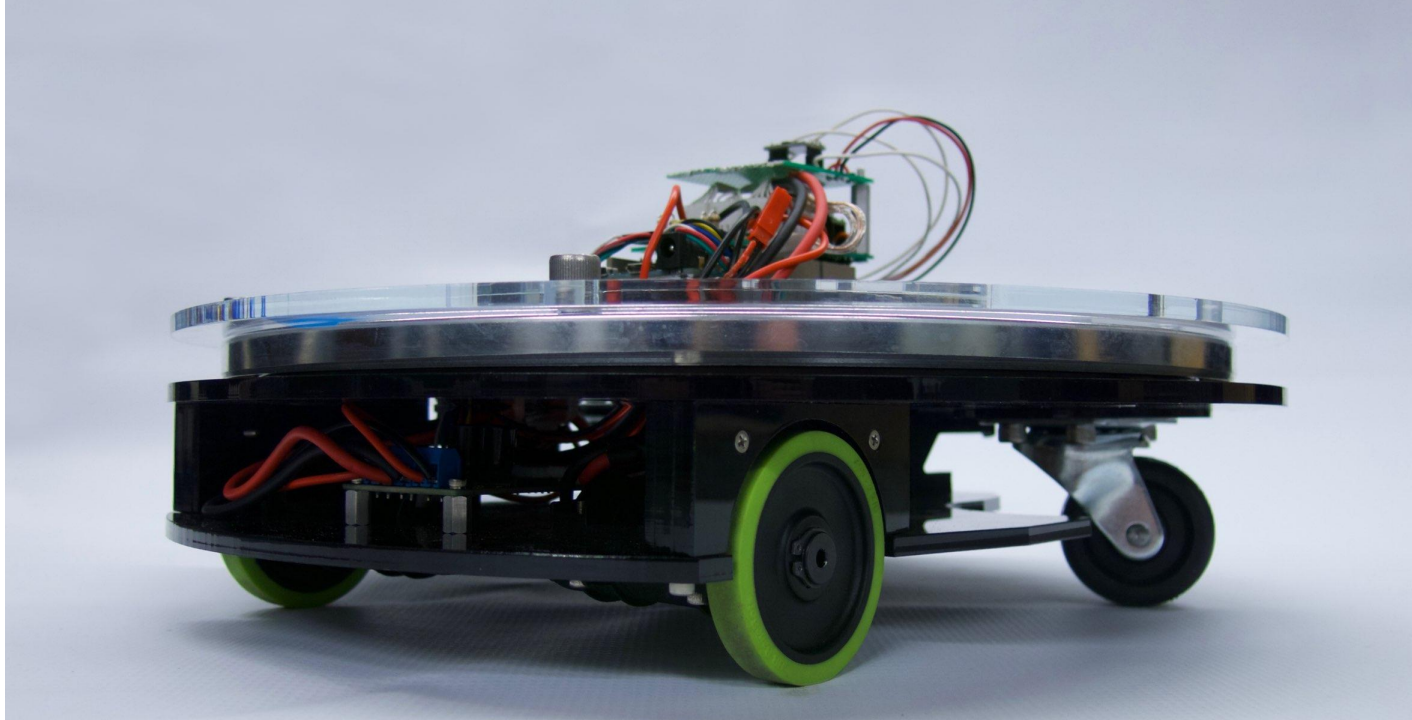
Fall 2025
Weekly Notes



Post-Summer

Update

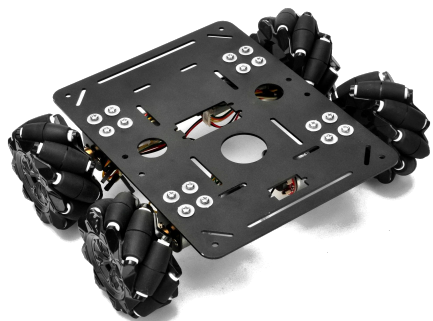
Original HAMR - ModLab



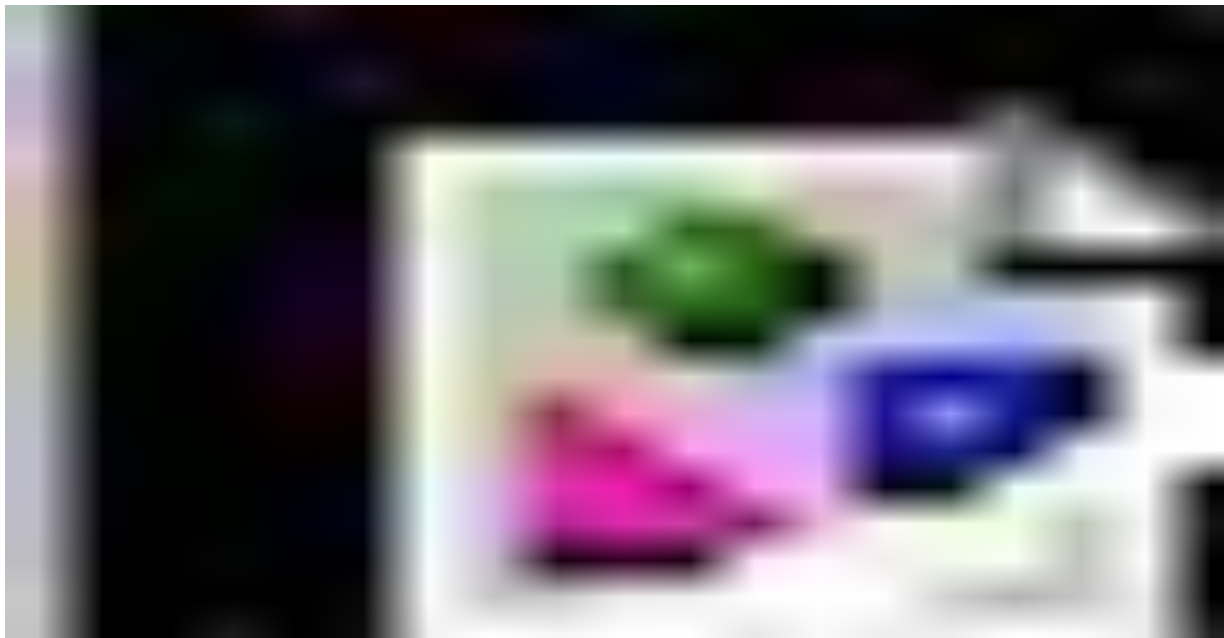
Original HAMR 2017

Holonomicity

The ability to reach any position and orientation in its configuration space regardless of its path.

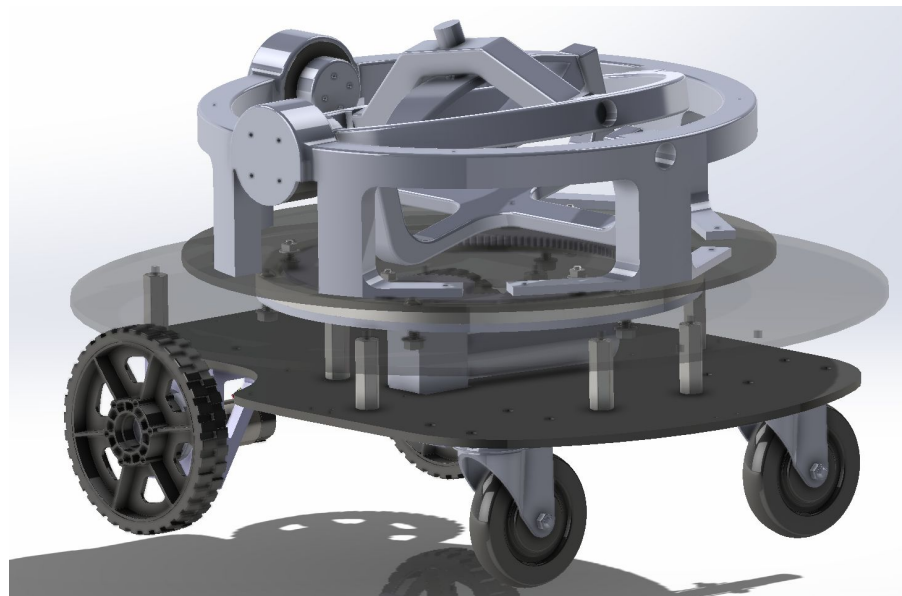
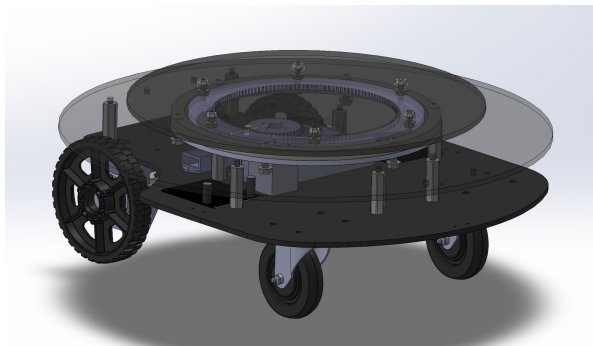


Mecanum Wheels



Visualizing Holonomic Motion

Hardware - ModLab



Hardware - ModLab

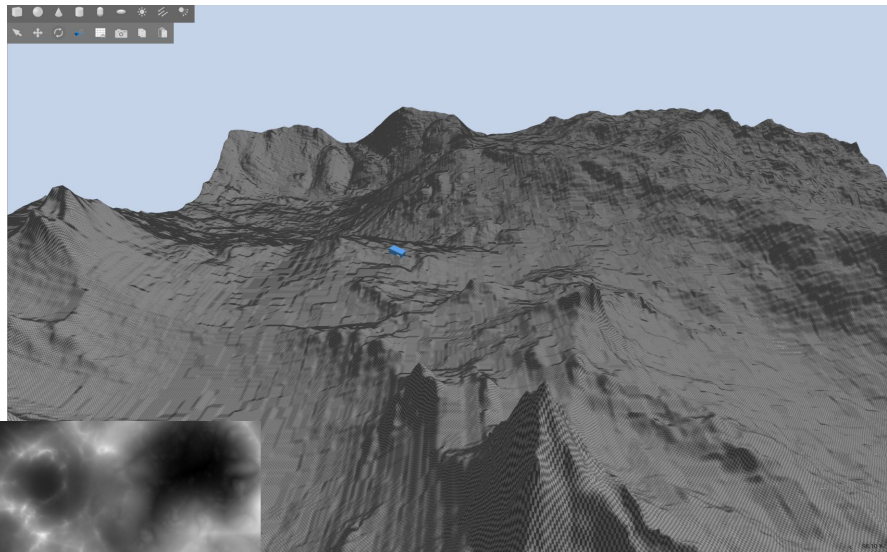


Holonomic Turret

COMPA - GMU

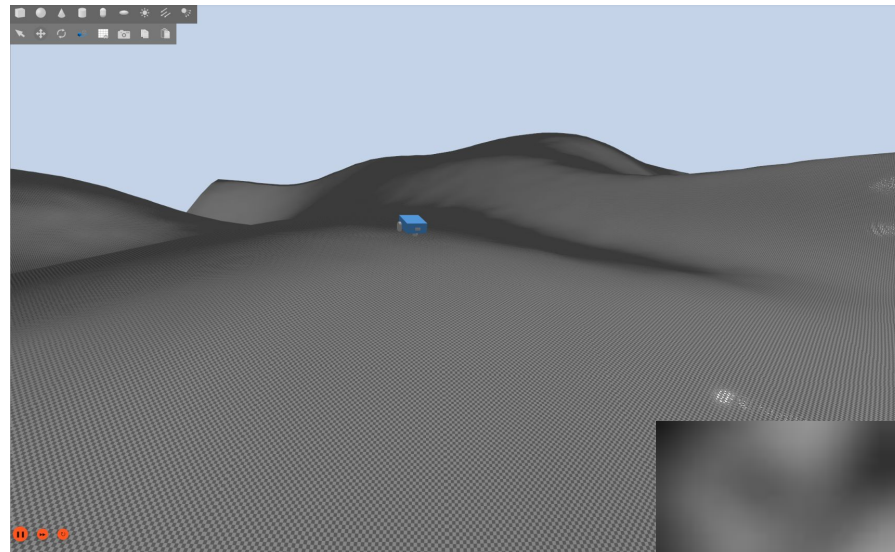
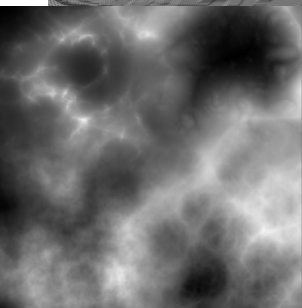


2.5D Terrain Gen - Simulation



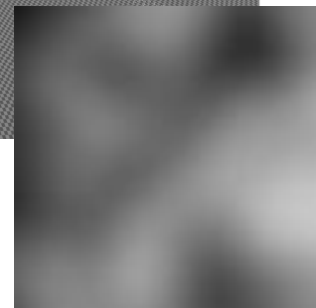
Rocky set up:

- 40mx40mx4m
- 1025x1025

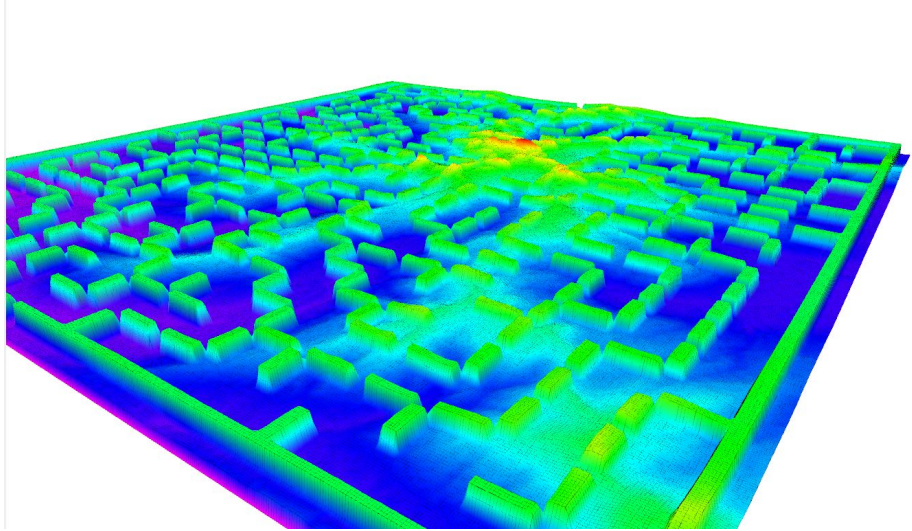
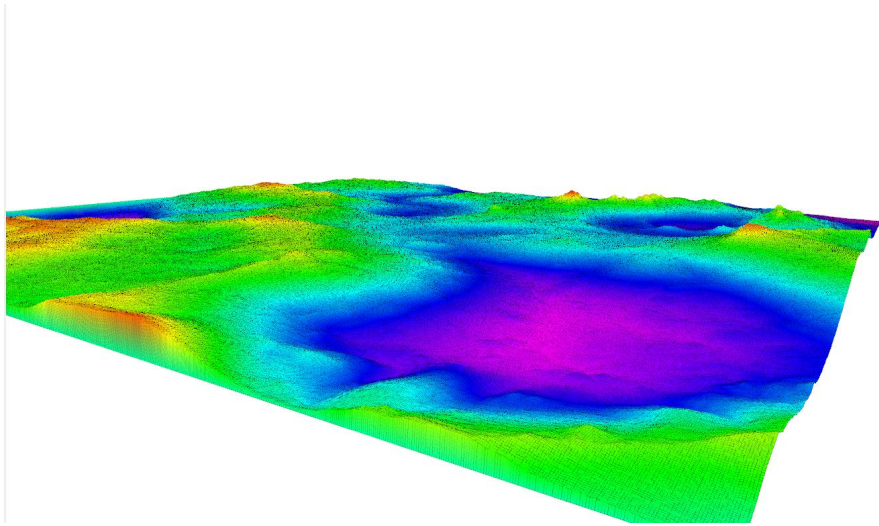


Smooth set up:

- 40mx40mx'10' m
- 257x257 + Gaussian blur 5px



2.5D Terrain Gen - Simulation



Elevation Map imported into Rviz

COMPA: Compensated Holonomic Off-Road Mobile Platform with Attitude Control

Anonymous Author(s)

Abstract—We present COMPA (Compensated holonomic Off-road Mobile Platform with Attitude control), an off-road differential-drive with offset turret platform that delivers holonomic control of a stabilized turret on uneven terrain. COMPA uses simple drive wheels on a rocker suspension to keep both wheels in contact with the ground, while an actively stabilized gimbal with three degrees of freedom, i.e., roll, pitch, and yaw, holds the turret at a commanded pose with respect to the world frame. The result is lateral, longitudinal, yaw, roll, and pitch motions that can be commanded independently at the turret, with the robot chassis free to absorb the terrain irregularities. We evaluate the physical COMPA and also in simulation on test courses and quantify its holonomicity using mobility ellipsoid. Experiments demonstrate holonomic turret motion and arbitrary trajectory execution without stop-to-reorient behavior as seen in robots with non-holonomic bases.

I. INTRODUCTION

Off-road robots must maneuver through uneven ground, surface discontinuities, deformable terrain, and varying friction. Their ability to navigate through these difficulties has facilitated their use in agriculture, disaster response, envi-



Fig. 1: COMPA demonstrating the ability to keep the turret horizontal (white plate) and directed (blue arrow), independent of the base pose, during holonomic off-road navigation.

1. **Hardware and Mechanical Design**
 - a. Elias Chavez, Parra Wang, Diego Salamanca
2. **Controls, Planning, and Simulation**
 - a. Cedric Hollande, Kartik Virmani
3. **Computer Vision**
 - a. Gbemiga Oloyede
4. **Embedded System**
 - a. Kaifan Yu



(09/20)

Weekly Updates

Semester Objectives

LONG TERM (Semester goal):

Goal:

- Build HAMR 3.0, similar to GMU's COMPA that is holonomic and has better off-road capabilities than HAMR 2.0
- Optimize holonomic motion planning for smooth off-road traversal over uneven terrain

Timeline: Brainstorming ideas, improving current system (sub-teams separately), integrating systems together, testing on vicon and then off-road

SHORT TERM:

- Set up Whole team organization
 - Meeting times, Pennovation meeting times
- Set up Sub-team organization
 - Meeting times (sub-team meetings)
 - Brainstorming for Semester Plan
- Onboarding done for everyone (preferably by Tuesday 09/23)

Hardware Next Steps

LONG TERM (Semester goal):

- **Goal:** Build a robot similar to GMU's COMPA that is holonomic and good off-road
- **Timeline:** Brainstorming ideas, CADDing and improving current base, Manufacturing

SHORT TERM:

- Get Familiar with our [HAMR SolidWorks Assembly](#)
- Checkout COMPA SolidWorks Assembly (STEP files)
- Brainstorm potential improvements on our system + GMU side and find best combination options
- Research potential mechanisms like rocker bogie that would work well for off-road holonomy
- **(Optional)** ROS 2 VM set up + simulation set up

Embedded Next Steps

LONG TERM (Semester goal):

- **Goal:** Integrate electronics to build a robot similar to GMU's COMPA that is holonomic and good off-road.
- **Timeline:** Brainstorming; Finding parts and integrating realsense, IMU and current sensors; Kalman filters and robust system integration

SHORT TERM:

- Get Familiar with our [HAMR SolidWorks Assembly](#) and [Embedded Kartik repo](#)
- Checkout COMPA SolidWorks Assembly (STEP files)
- Brainstorm potential improvements on our system + GMU side and find best combination options
- Research potential sensors and improvements
- (Optional) ROS 2 VM set up + simulation set up

LONG TERM (Semester goal):

- **Goal:** Integrate Local SLAM to get an understanding of the 2.5D terrain + subject tracking
- **Timeline:** Brainstorming and researching algorithms; Simulation Integration; Real robot + World Integration

SHORT TERM:

- Simulation set up
- Algorithms research
- Planning for progressive integration



(09/26)

Weekly Updates

Simulation and Planning Next Steps

LONG TERM (Semester goal):

- Optimization on path planning
- Add autonomy / intelligence through ROS2 Actions and Services + more? -> Simulate a mission, add a Lidar to showcase some stuff?
- Convert Jacobian Controller to C++?

SHORT TERM:

Same as before:

- 2.5D Planner:
- A* Cost adaptation (Cedric & Kartik)
- Policy Iteration ? (Kartik)
- Add a moving target and try to mimic CV Algorithm (prepare the bridge)
- Caster improve in sim and add to COMPA

Hardware Next Steps

LONG TERM (Semester goal):

- **Goal:** Build a robot similar to GMU's COMPA that is holonomic and good off-road
- **Timeline:** Brainstorming ideas, CADDing and improving current base, Manufacturing

SHORT TERM:

- Offroad mobile base
 - Rocker mechanism (similar to COMPA)
 - Bigger wheels, order any extra necessary parts (TPU in-house design)
 - Remake frame with rails, less acrylic, protect electronics - enclosed?
- Gimbal
- Caster Design
 - Spherical
 - Deformable wheel
 - Actuated?

Embedded Next Steps

LONG TERM (Semester goal):

- **Goal:** Integrate electronics to build a robot similar to GMU's COMPA that is holonomic and good off-road.
- **Timeline:** Brainstorming; Finding parts and integrating realsense, IMU and current sensors; Kalman filters and robust system integration

SHORT TERM:

- Kartik: RPi set up and odometry improvements to test on the vicon

LONG TERM (Semester goal):

- **Goal:** Integrate Local SLAM to get an understanding of the 2.5D terrain + subject tracking
- **Timeline:** Brainstorming and researching algorithms; Simulation Integration; Real robot + World Integration

SHORT TERM:

- Simulation set up
- Algorithms research & figure out how to improve them for our use case
- Draw out a full pipeline for how the vision stack would work and interact with other components of the project
- Before implementing YOLO, create a fake target that publishes bouncing boxes for a colored object. Integrate a real detection algorithm after using RGB frames.
- Planning for progressive integration



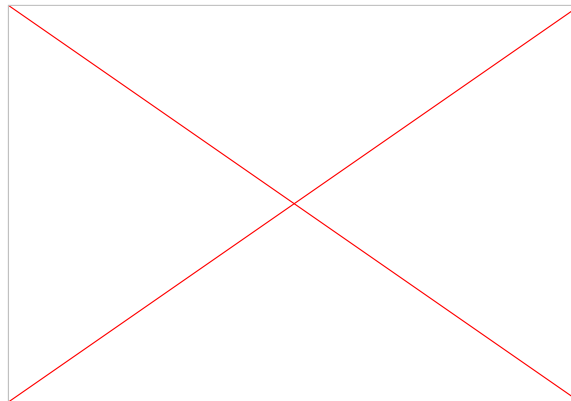
(10/03)

Weekly Updates

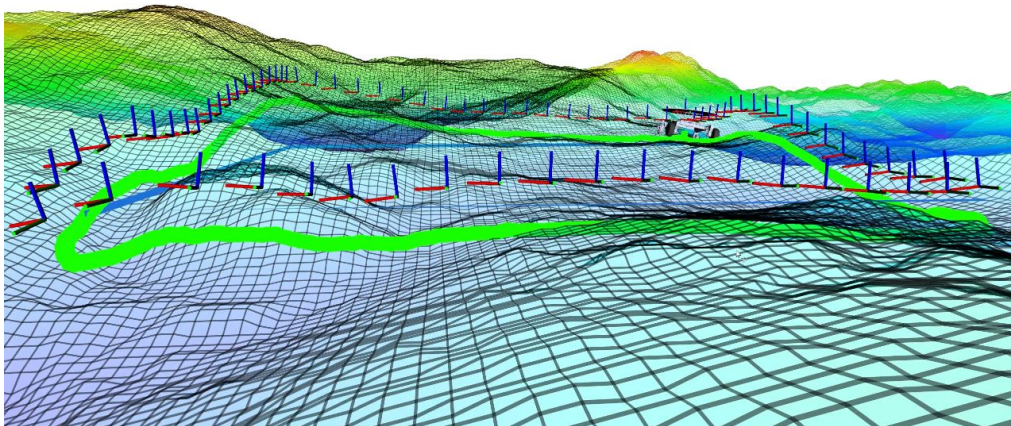
Simulation: Updates

This week:

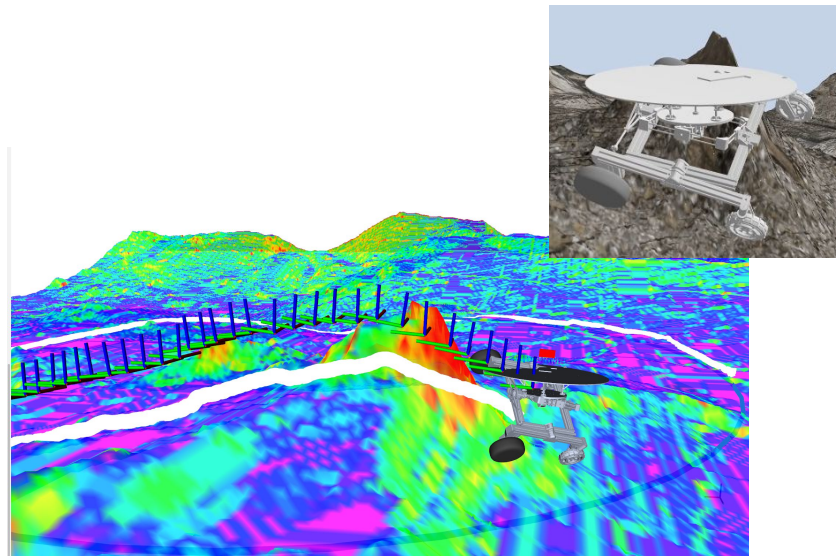
1. Submitted ICRA Video
 - a. Implemented Gimbal Control + gimbal performance showcase (tf graphing in Rviz)
2. Implemented Rviz visualization of terrain traversability
3. Simulation->Hardware: Vicon Testing



Simulation: Gimbal Performance



Mapping TFs throughout the trajectory

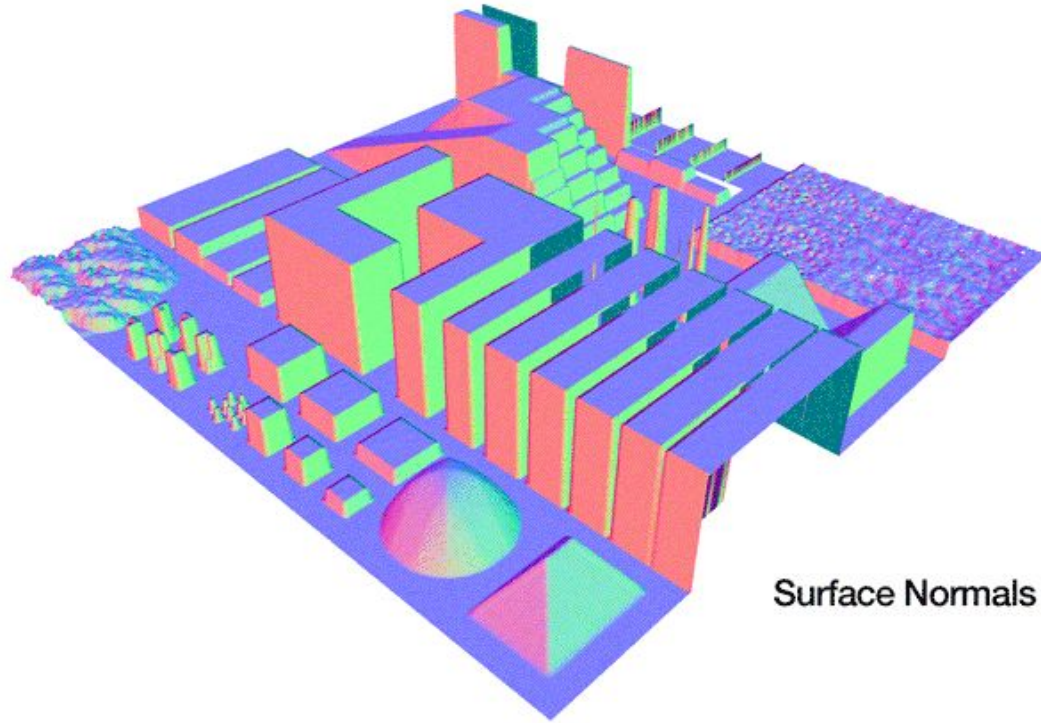


Pushing the limits!

Notes from the experiments:

- The base link could be higher to avoid “grounding out”
- Turret plate could be placed higher (or reduce radius) to avoid self-colliding
- For stable camera feedback it might be helpful to have a physical damper
- Adding a feedforward based on 2.5d terrain could also help reduce noise for roll and pitch

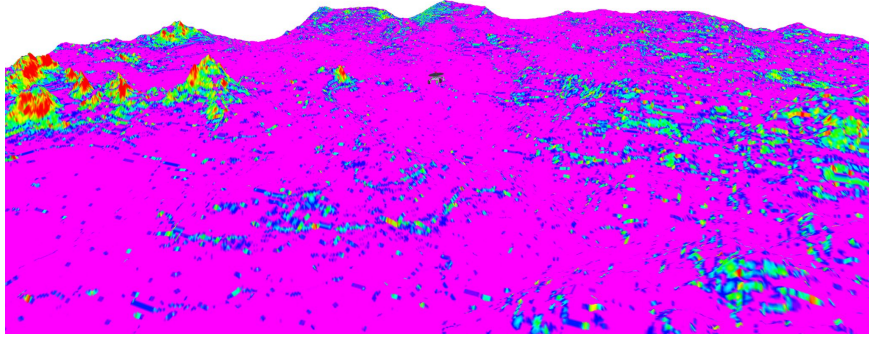
Simulation: 2.5D Planning



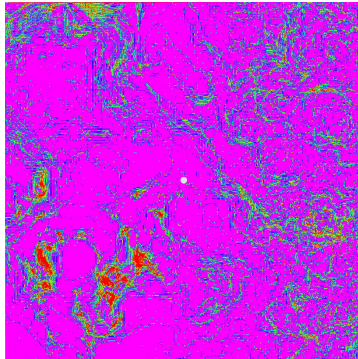
Surface Normals

Different filters from Grid Map

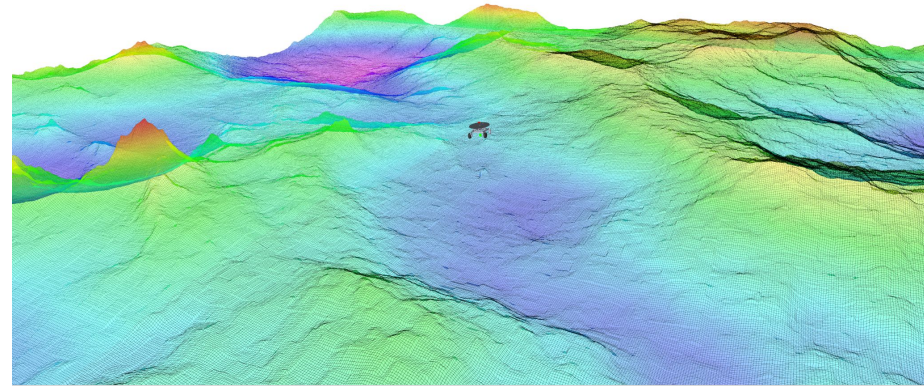
Simulation: 2.5D Traversability



Traversability Map (adapted from Gridmap pkg)

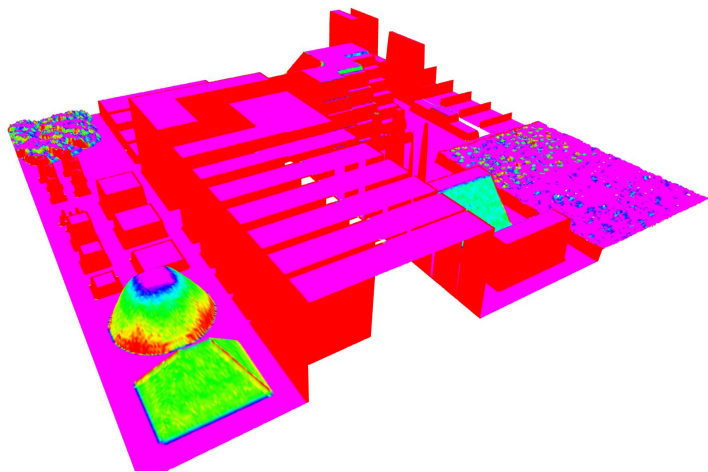


Traversability Map 2D

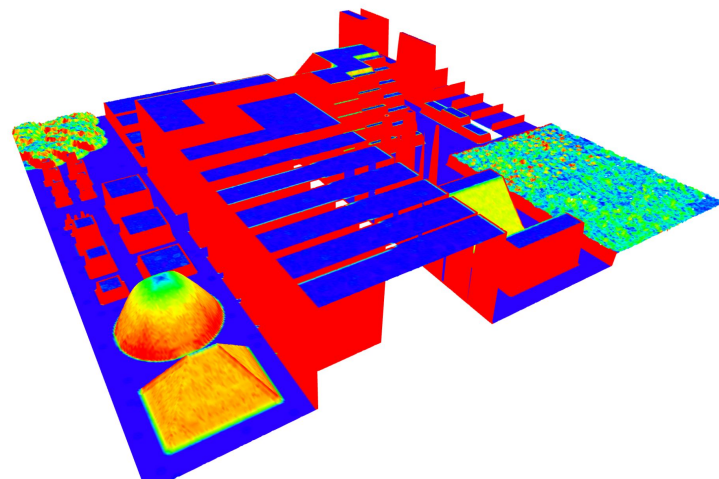


Regular Elevation Map

Simulation: 2.5D Traversability



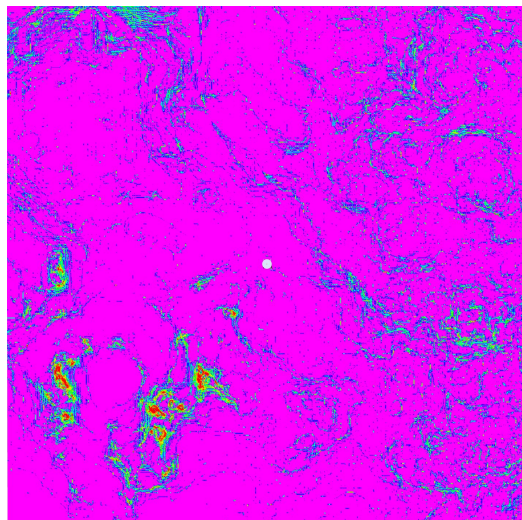
Max Intensity: 0.5



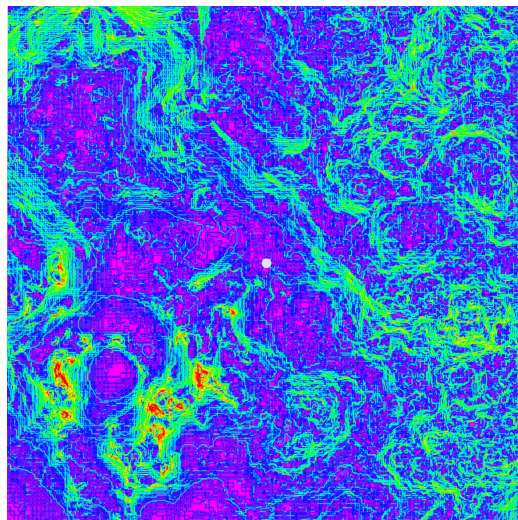
Max Intensity: 1.2

*Max Intensity Parameter inversely correlated to
Traversability Capabilities (from >0 to ~ 5.0)*

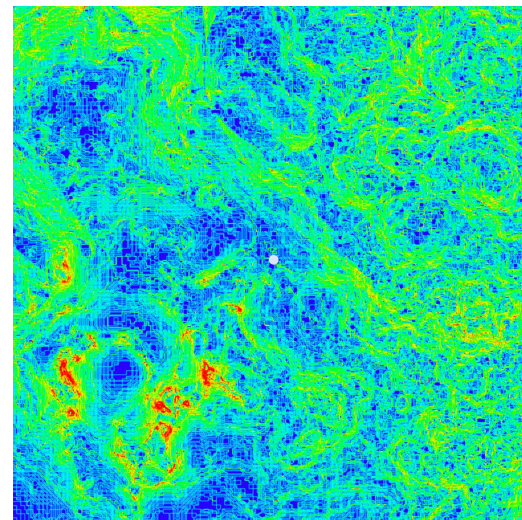
Simulation: 2.5D Traversability



Max Intensity: 0.5



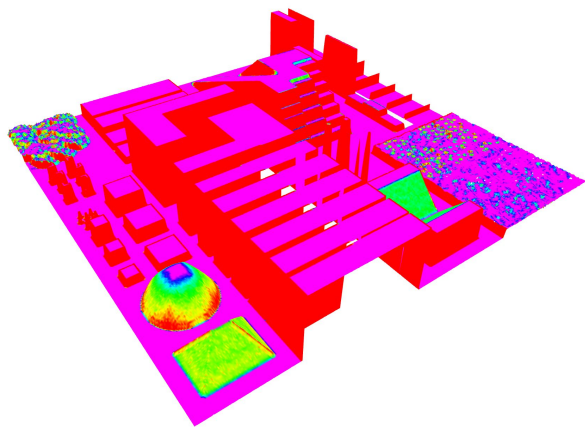
Max Intensity: 0.9



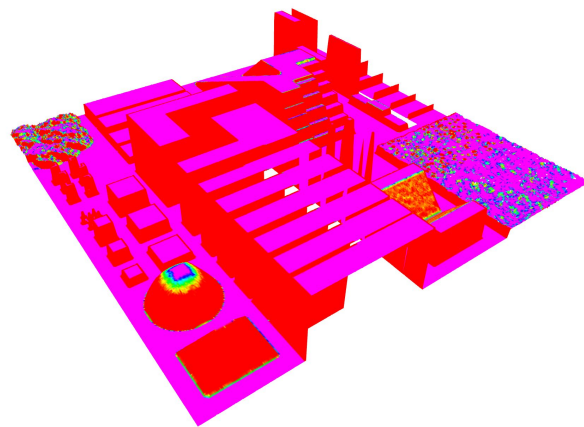
Max Intensity: 1.2

*Max Intensity Parameter inversely correlated to
Traversability Capabilities (from >0 to ~ 5.0)*

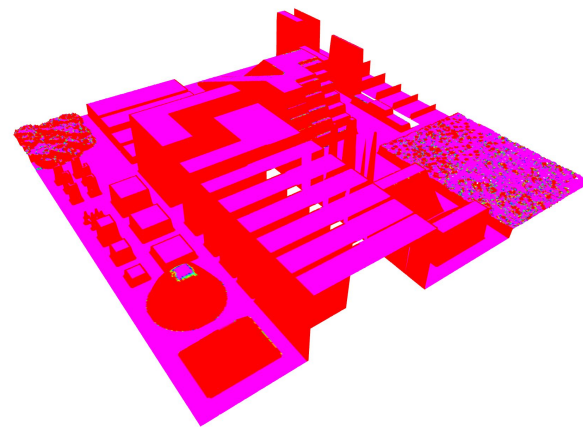
Simulation: 2.5D Traversability



Min, Max Intensity: 0.0, 0.6



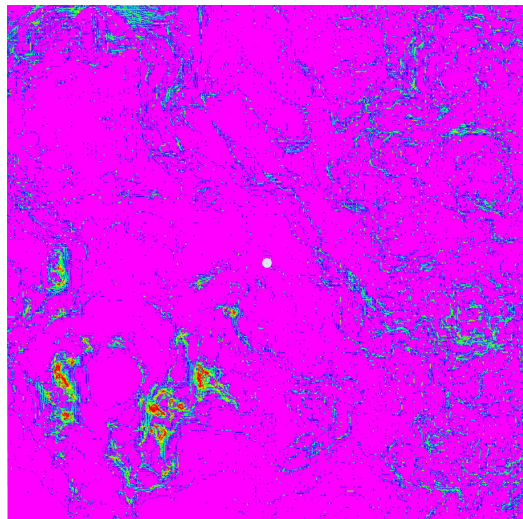
Min, Max Intensity: 0.25, 0.6



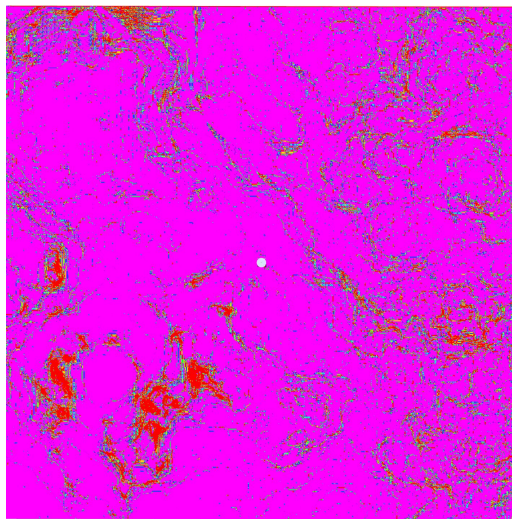
Min, Max Intensity: 0.5, 0.6

*Min Intensity Parameter correlated to Traversability
Sensitivity (from >0 to ~5.0)*

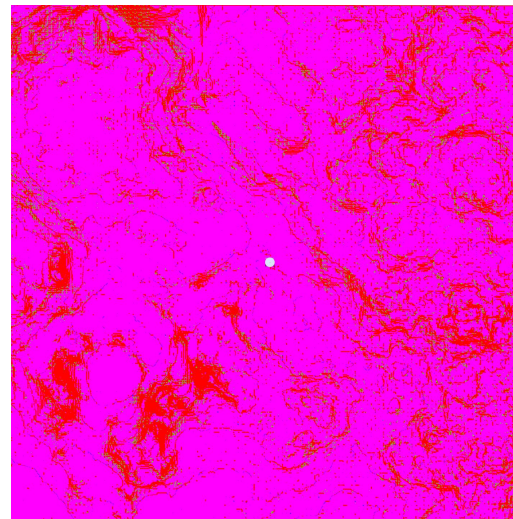
Simulation: 2.5D Traversability



Min, Max Intensity: 0.0, 0.5



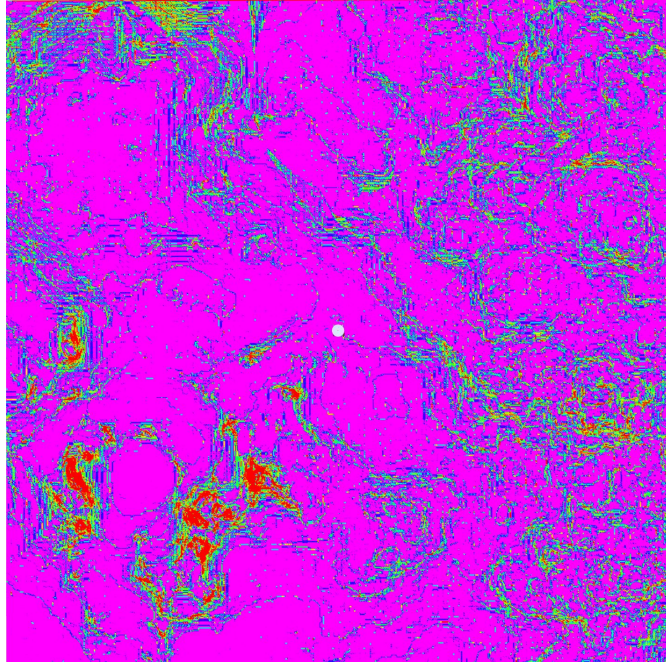
Min, Max Intensity: 0.3, 0.5



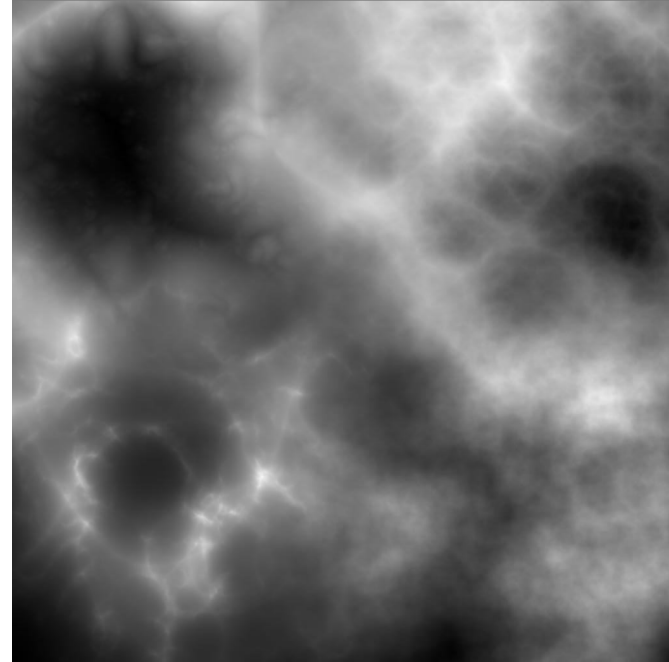
Min, Max Intensity: 0.45, 0.5

*Min Intensity Parameter correlated to Traversability
Sensitivity (from >0 to ~5.0)*

Simulation: 2.5D Traversability

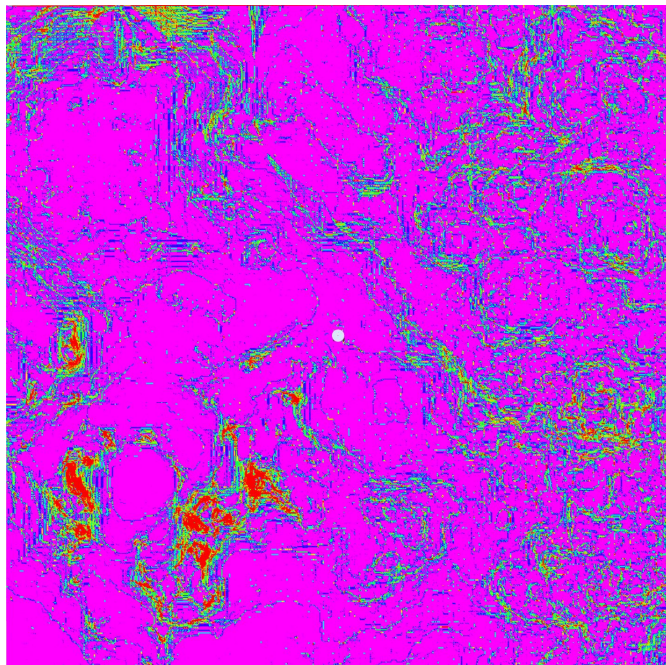


Tuned Traversability (Map: 0.2, 0.6)

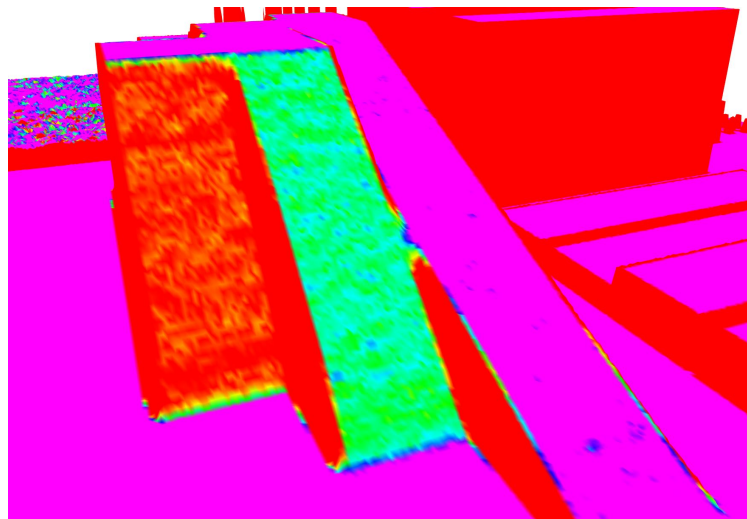


Corresponding Heightmap

Simulation: 2.5D Traversability

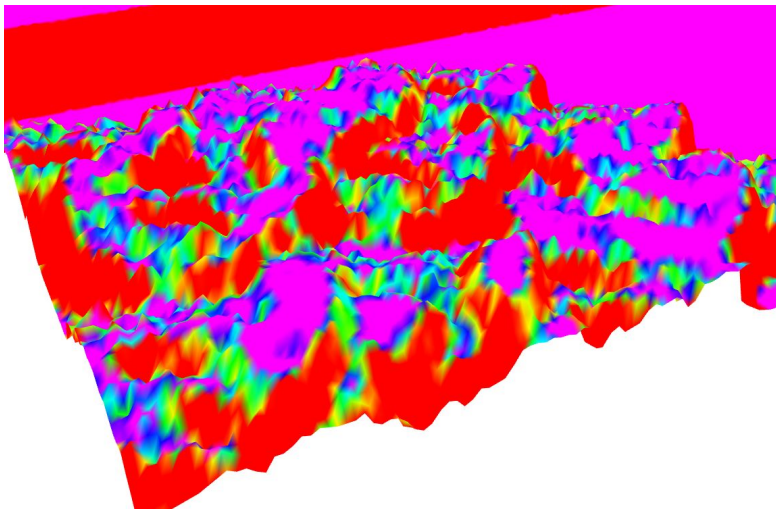


Tuned Traversability (Map: 0.2, 0.6)

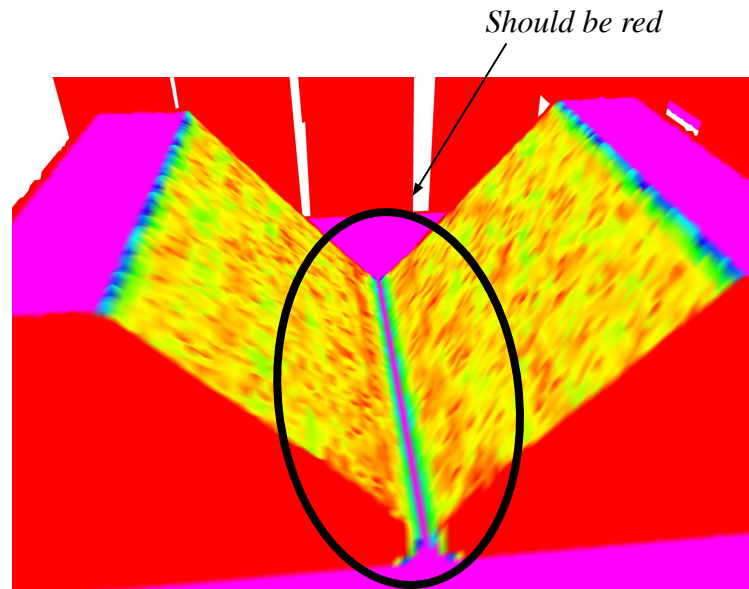


Corresponding Slope (Map: 0.2, 0.6)

Simulation: 2.5D Traversability

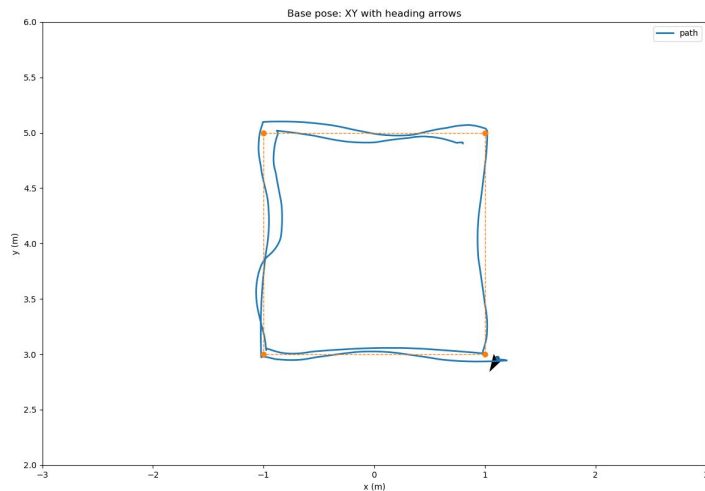


*Shouldn't be as much red - take into consideration **wheel** and **caster size***

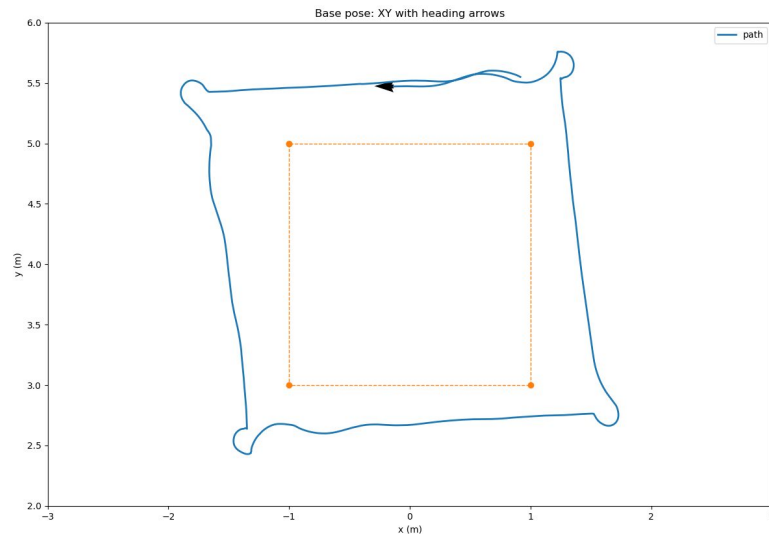


*Terrain **roughness** in regions larger than pixels not taken into account*

Simulation: Vicon Testing

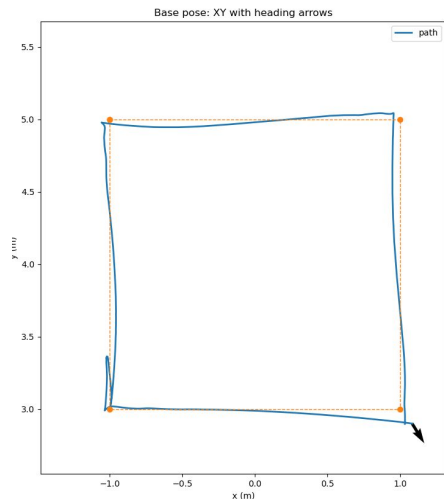


Back wheel drive auto (slow)

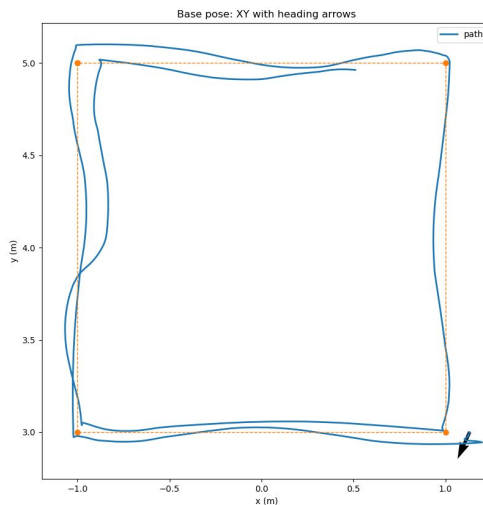


Front wheel drive auto (slow)

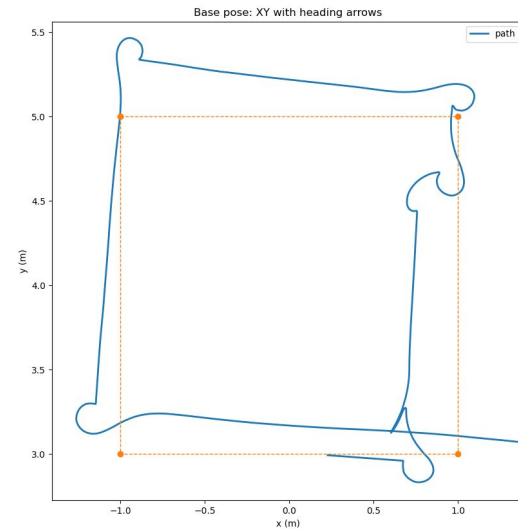
Simulation: Vicon Testing



Back wheel drive manual (fast)

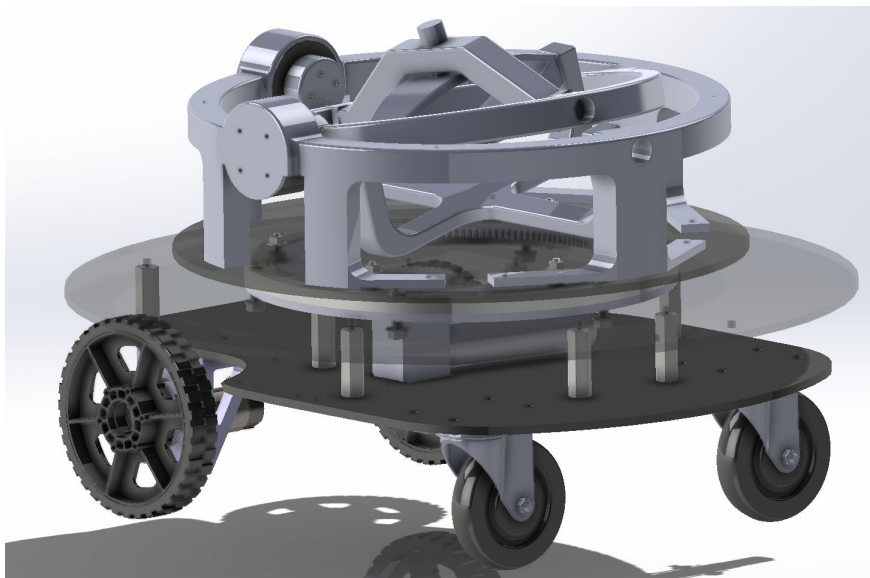


Back wheel drive manual (slow)

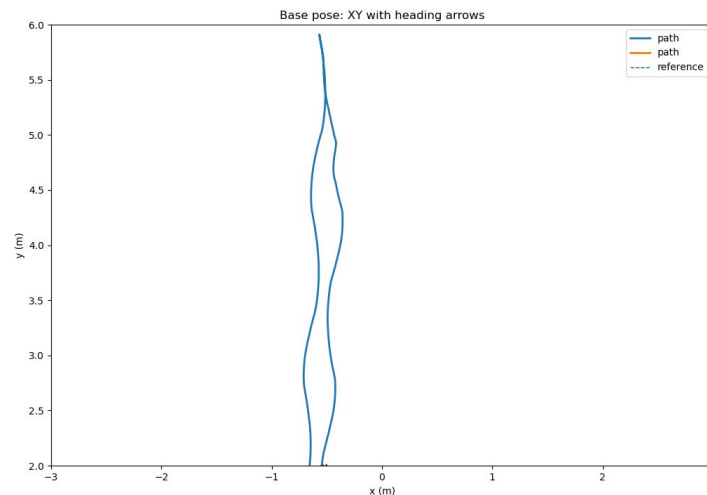


Front wheel drive manual (normal)

Simulation: Vicon Testing



Front swivel caster limitations



*Trying to stay in a straight line (with our
holonomic closed-loop controller)*

Simulation and Planning Next Steps

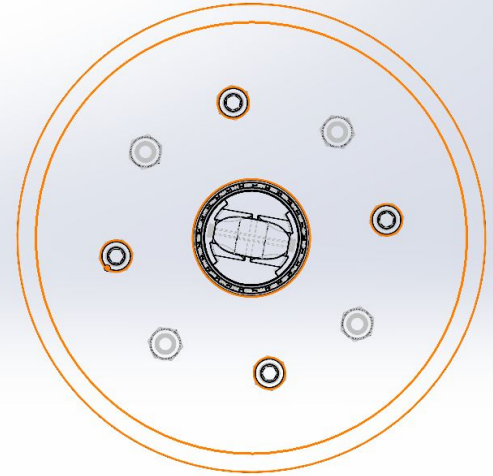
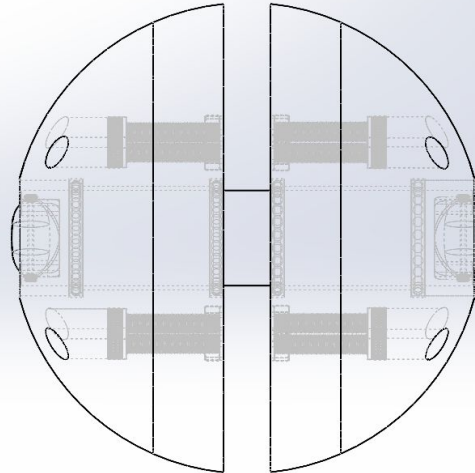
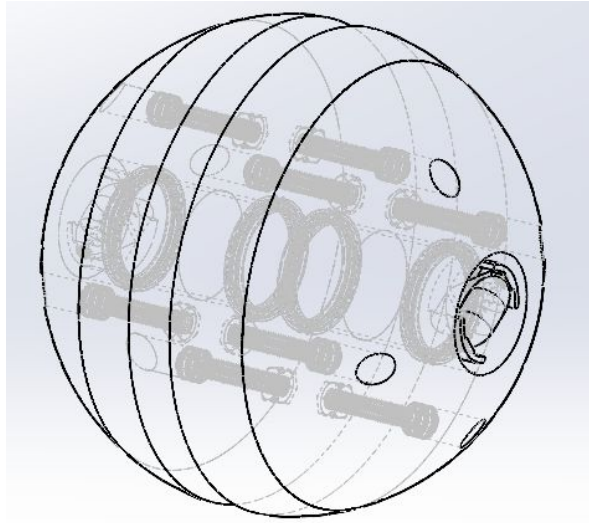
LONG TERM (Semester goal):

- Optimization on path planning
 - Create a costmap to optimize planning and controls -> non-linear controller
 - Based on dynamics of our system (rocker mechanism?) - goal: choose path that will stabilize gimbal the best
- Add autonomy / intelligence through ROS2:
 - Actions and Services
 - Components and lifecycle nodes for faster compute and Hardware Integration
 - Simulate a mission, add a Lidar to showcase some stuff?
- Convert Jacobian Controller to C++?
- Improve smoothness of gimbal - add feedforward to

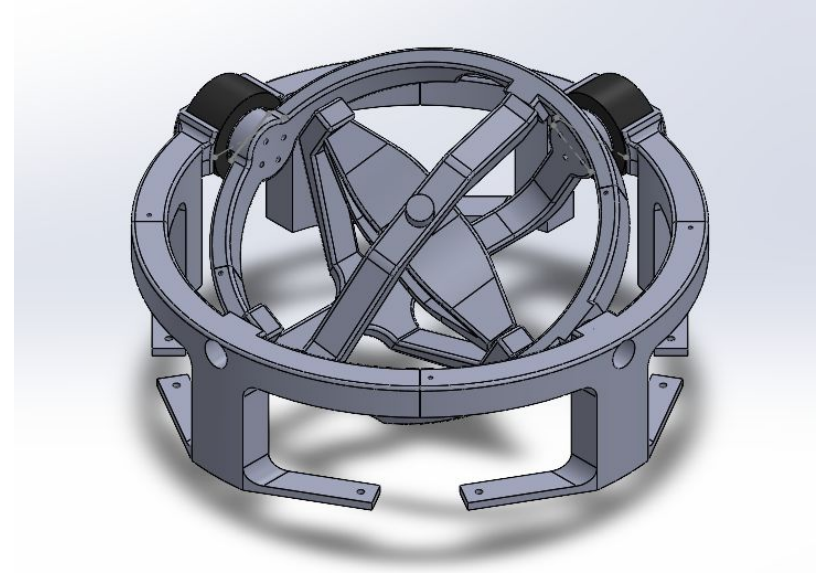
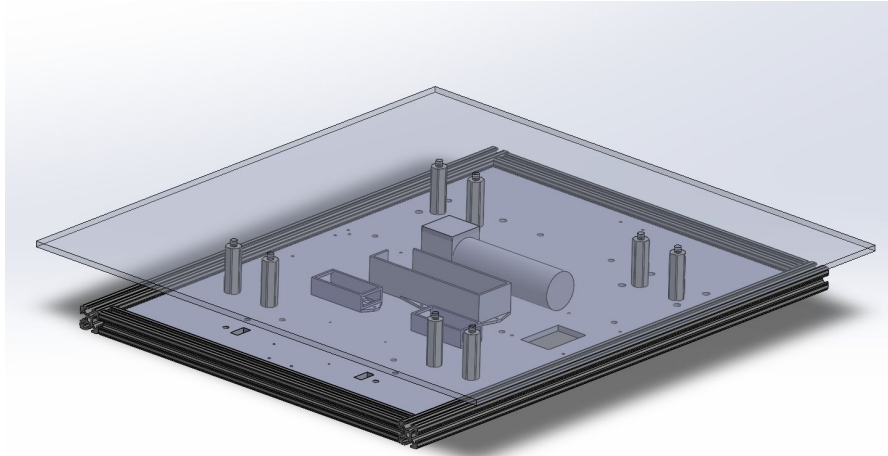
SHORT TERM:

- 2.5D Planner:
 - Traversability Paper + Math
 - A* Cost adaptation (Cedric & Kartik) - Later used to compare to our optimization algo
 - Policy Iteration ? (Kartik)
- Add a moving target and try to mimic CV Algorithm (prepare the bridge)
- Caster improve in sim and add to COMPA

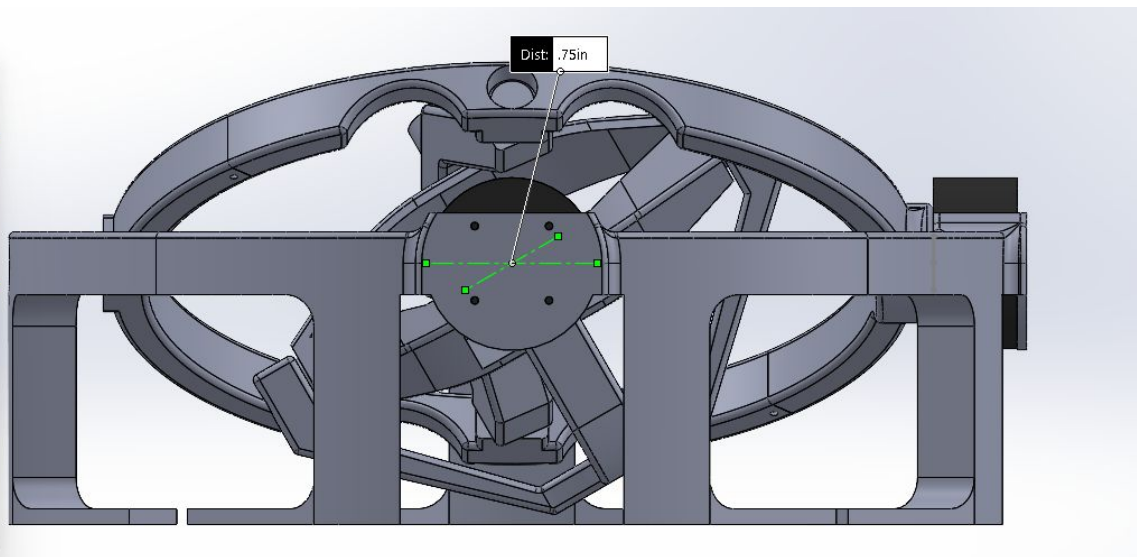
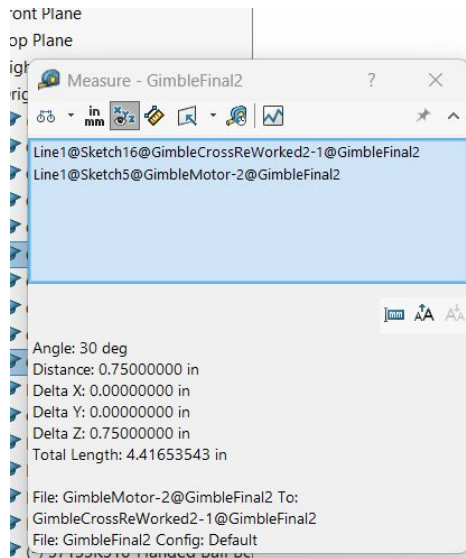
Hardware: Sphere Caster



Hardware: Base Redesign

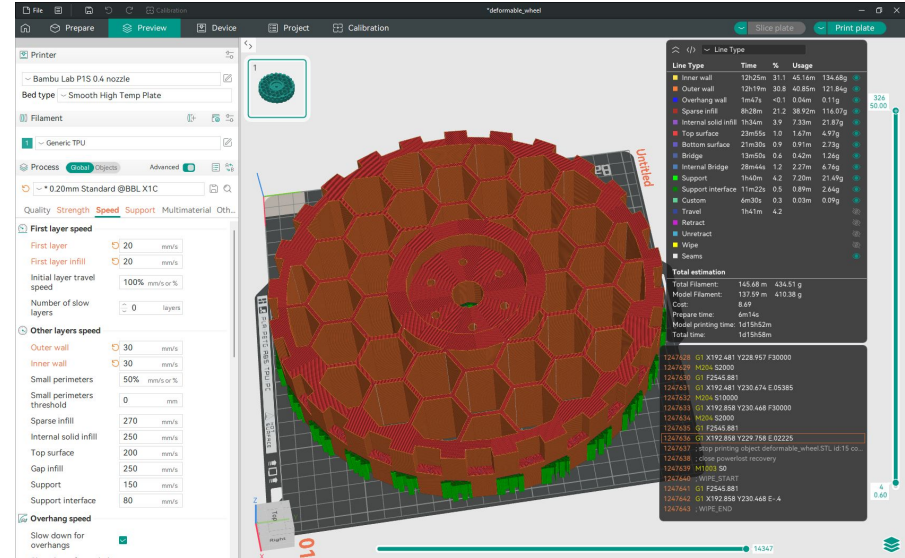
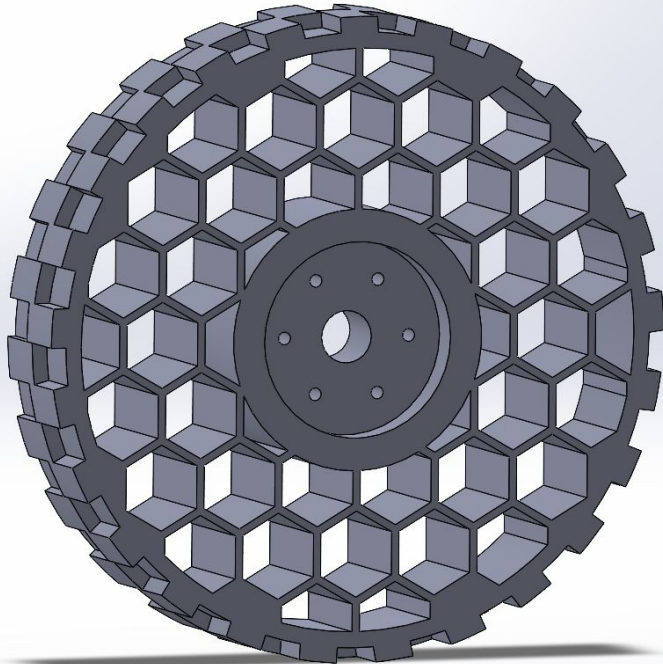


Hardware: Base Redesign



Increased gimbal ROM to 30 deg

Hardware: Deformable Wheels?





(10/10)

Weekly Updates

Simulation and Planning Next Steps

LONG TERM (Semester goal):

- Optimization on path planning
 - Create a costmap to optimize planning and controls -> non-linear controller
 - Based on dynamics of our system (rocker mechanism?) - goal: choose path that will stabilize gimbal the best
- Add autonomy / intelligence through ROS2:
 - Actions and Services
 - Components and lifecycle nodes for faster compute and Hardware Integration
 - Simulate a mission, add a Lidar to showcase some stuff?
- Convert Jacobian Controller to C++?
- Improve smoothness of gimbal - add feedforward to

SHORT TERM:

- 2.5D Planner:
 - READ Traversability Paper + Math
 - A* Cost adaptation (Cedric & Kartik) - Later used to compare to our optimization algo
 - Policy Iteration ? (Kartik)
- Vicon and real robot Autonomous Control
- Add a moving target and try to mimic CV Algorithm (prepare the bridge)
- Caster improve in sim and add to COMPA



(10/22)

Weekly Updates

— OLD Slide —

Goal:

1. Create an autonomous algorithm which minimizes the disturbance caused by navigating **Rough Terrain** (forrests, rocky planes)
2. Thus optimizing for **holonomic behavior** throughout trajectories, while maintaining the **safest** and most **energy efficient** motion planning for HAMR

Current Literature:

1. Focuses in 2.5D Traversability Planning for DD or Ackerman Steering Robots
2. 2 Main Strategies (for traversability analysis):
 - a. Classic Geometric approaches
 - b. Learning approaches
- 3.

Goal:

Develop a 2.5D, orientation-aware traversability optimizer that plans SE(2) trajectories minimizing **terrain-induced disturbance** and **energy use**, while satisfying stability and collision constraints.

The planner must exploit **holonomic freedom** (free heading choice at each step) to align motion with **direction-dependent** terrain costs (slope, roughness, edges) for HAMR.

- **Disturbance:** quantify via body accelerations, jerk, roll/pitch rates, contact shocks
- **Energy:** mechanical work per meter, estimated from uphill/downhill/lateral slip and terrain roughness based on HAMR dynamics

Current Literature (ChatGPT Summarized):

1. Geometry-only (2.5D)
 - a. Slope and roughness layers; multi-layer cost maps
 - b. Edge/step detection; curvature-based penalties
 - c. Anisotropic metrics on Digital Elevations Maps (DEMs)
 - d. Heuristics: slope-aware or Hybrid A*; direction-dependent motion primitives
2. Physics/energy-informed
 - a. Grade + rolling resistance models for energy per meter
 - b. Side-slope stability; tip-over metric; traction limits from friction cones
 - c. Disturbance proxies: predicted body accel from terrain PSD and speed
3. Uncertainty-aware
 - a. Elevation variance from sensor fusion; conservative inflation
 - b. Risk measures (CVaR) in cost; chance constraints for stability
4. Learning-based
 - a. CNNs on heightmaps predicting per-cell traversability or cost-to-go
 - b. Self-supervised labels from vehicle telemetry (slip, current, IMU jerk)
 - c. Orientation-aware networks: input current heading; output $c(x,y,\psi)$ or $v_{\max}(x,y,\psi)$
 - d. Sim-to-real with domain randomization on DEM statistics; few-shot finetune
5. Hybrid
 - a. Learned priors for cost layers or heuristics; classical search for guarantees
 - b. Bilevel: planner proposes path, learned module sets speeds or refines heading

Holonomic Traversability Optimization

20-30 [Papers to read](#) (currently at ~10)

	A	B	C	D	E	F
1	Stage	Paper	Type	Description	Importance	Notes
2	1. Foundations & Key Concepts	[2294, 10883] A Survey of Traversability Estimation for Mobile Robots	2.5D + Traversability	Excellent overview of terrain traversability methods using geometry (slope/roughness), physics (stability), learning (CNNs/RL), and semantics. Frames the full problem and terminology.	Read Conceptual parts	- Paper emphasizes learning-based traversability estimation (ML → Deep Learning). - Historical progression: geometric methods → handcrafted ML features → deep learning. - Geometry-only approaches are reliable but limited for: - deformable terrain (mud, sand, snow) - low-friction/wet surfaces - high-speed dynamics (slip not modeled by elevation alone) - occlusions/incomplete elevation maps - domain shift (appearance changes across environments) <i>"Geometry-based traversability methods remain competitive and interpretable but struggle in unstructured environments with deformable soils, slip-dominated dynamics, and domain shifts, motivating hybrid approaches that integrate semantics or learned terrain priors [A Survey of Traversability Estimation for Mobile Robots, 2022]."</i>
3		Practical Search Techniques in Path Planning for Autonomous Driving	Hybrid A* + AGV Planni...		Skim	Hybrid A* doesn't consider holonomic robots, but takes into consideration kinematics of car and continuous inside each cell (we don't take the center of the cell like in regular A*)
4		High-Efficiency Navigation of Nonholonomic Mobile Robots Based on Improved Hybrid A* Algorithm	Hybrid A* + AGV Planni...		Read	Hybrid A* + mpcc for non-holonomic ackerman systems - Hybrid A* is lacking traversability gauge although it initiates considering kinematic constraints
5		Energy-Optimal Ground Vehicle Trajectory Planning on Deformable Terrain	Hybrid A* + AGV Planni...		Skim	
6		Traversability Analysis and Path Planning for Autonomous Wheeled Vehicles on Rugged Terrain	2.5D + Traversability	This paper wants to plan safe and efficient paths for an autonomous wheeled vehicle driving on rigid terrain (like compacted soil, mining roads, concrete, asphalt - not deformable mud or sand). They argue that many papers focus only on perception (vision/LIDAR) to estimate traversability, but don't consider how the vehicle itself behaves - things like speed, friction, tire pressure, vibration, and vehicle load. This paper tries to fix that.	Must Read	TV-Hybrid A* planner: introducing the consideration of terrain characteristics into the Hybrid A* planner (and vehicle limitations) to create traversability boundaries determined by a three-layer neural network model.
7	2. Core 2.5D Traversability	A Universal Grid Map Library: Implementation and Use Case for Rough Terrain Navigation	2.5D + Traversability	Introduces how to map a discrete Gridmap in Rviz with filters such as normal vectors, slope, roughness and traversability based off a 2.5D heightmap.	Must Read	gridmap library used in rviz and ros2
8		Rapid Traversability Assessment in 2.5D Grid-based Map on Rough Terrain	2.5D + Traversability	Introduces cost based on slope + roughness per cell. Foundational 2.5D grid-based traversability.	Skim	
9		Path planning for mobile robot in a 2.5-dimensional grid-based map	2.5D + Traversability	Straightforward expansion of 2.5D planning, introduces elevation gradients and rolling danger. Good early baseline.	Skim if time	
10		3D traversability analysis and path planning based on mechanical effort for UGVs in forest environments - ScienceDirect	2.5D + Traversability	Introduces mechanical energy effort as a cost rather than just morphology. Very relevant to our "effort-aware" cost idea.	Read	We can have 2 ways to estimate the roughness of the terrain (solely on point clouds such as lidar or learning-based by predicting IMU output from geometric point cloud data w/ NN); this paper compares both
11		Traversability analysis for off-road environments using locomotion experiments and earth observation data - ScienceDirect	2.5D + Traversability	Connects real robot motion with terrain statistics. Strong validation of field-based cost design.	Good Read	
12				Terrain pose mapping to describe impact of terrain on robot. With this mapping, we obtain the SE(3) state of the robot on uneven terrain for a given state in SE(2). Then, based on it, we present a trajectory optimization		Cool video to IROS 2023:

Holonomic Traversability Optimization

1

[\[2204.10883\] A Survey of Traversability Estimation for Mobile Robots](#)

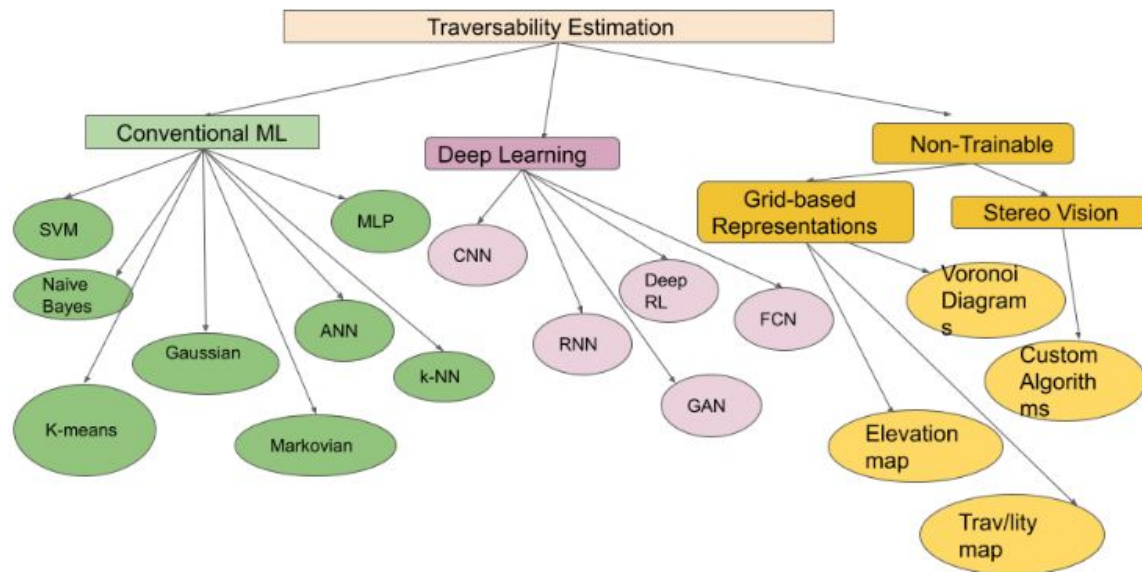


Figure 1: Overview of the different techniques discussed in this survey

Holonomic Traverability Optimization

2/3 Hybrid A* (Non-Holonomic and 2D):

1. [Practical Search Techniques in Path Planning for Autonomous Driving](#)
2. [High-Efficiency Navigation of Nonholonomic Mobile Robots Based on Improved Hybrid A* Algorithm](#)

Considers kinematics, and ‘continuous’ planning: motion primitives and nodes can be anywhere within each cell

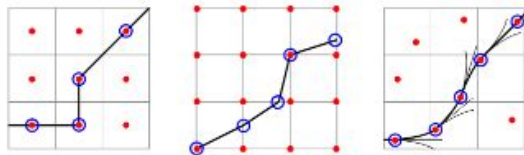
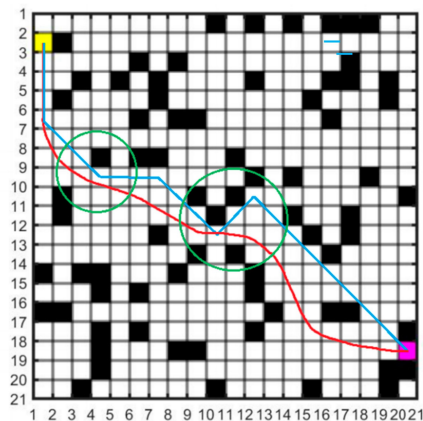
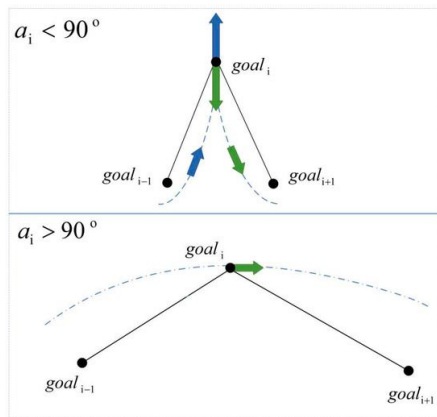


Figure 2: Graphical comparison of search algorithms. Left: A* associates costs with centers of cells and only visits states that correspond to grid-cell centers. Center: Field D* (Ferguson and Stentz 2005) and Theta* (Nash et al. 2007) associate costs with cell corners and allow arbitrary linear paths from cell to cell. Right: Hybrid A* associates a continuous state with each cell and the score of the cell is the cost of its associated continuous state.



— Conventional algorithm
— Improved algorithm
■ Starting point
■ End point

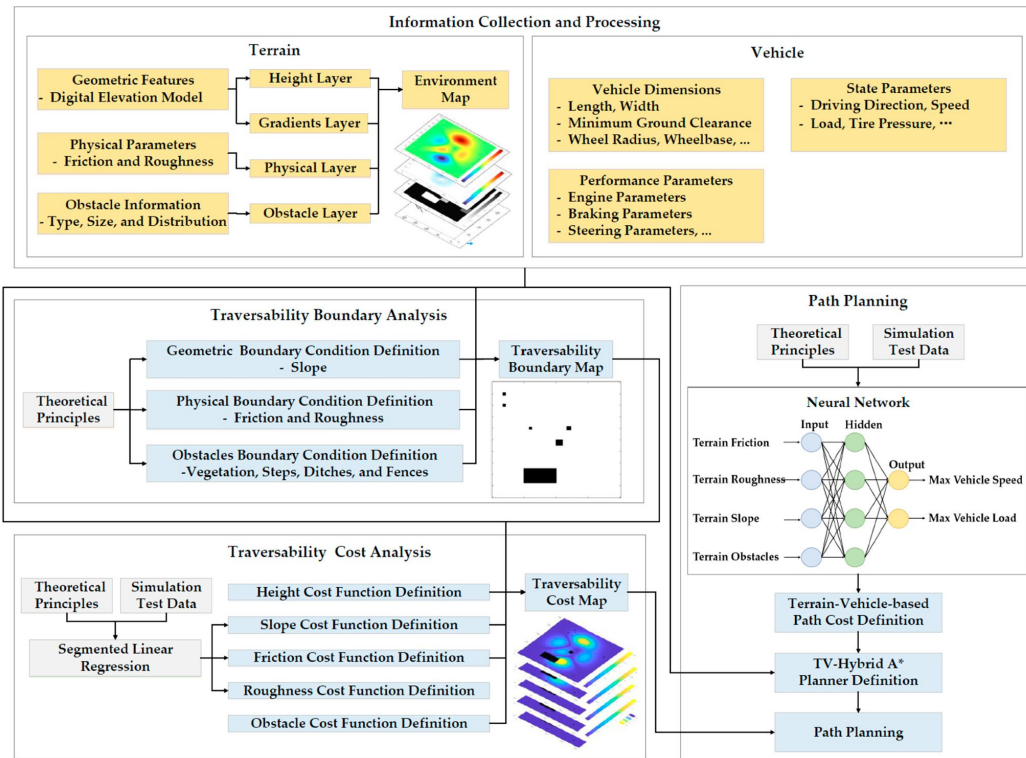


Holonomic Traversability Optimization

4

Traversability Analysis and Path Planning for Autonomous Wheeled Vehicles on Rigid Terrains:

Non-Holonomic Hybrid approach combining classic geometric, physical parameters and physics analysis reinforced by a Neural Network for Traversability Cost Analysis



Holonomic Traverability Optimization

4 Traversability Analysis and Path Planning for Autonomous Wheeled Vehicles on Rigid Terrains

Segmented Linear Regression for Traversal Cost (slope, friction roughness...):

$$\text{Cost}_{\text{slope}} = \begin{cases} a_1 \cdot \theta + b_1 & \text{if } 0^\circ \leq \theta \leq 5^\circ \\ a_2 \cdot \theta + b_2 & \text{if } 5^\circ < \theta \leq 15^\circ \\ a_3 \cdot \theta + b_3 & \text{if } \theta > 15^\circ \end{cases}$$

Max acceleration/deceleration without slipping

$$\varphi_L = \min \{ \varphi_{L_{\text{acc}}}, \varphi_{L_{\text{dec}}} \}$$

$$\varphi_{L_{\text{acc}}} = \frac{T_{tq} \cdot i}{N \cdot r}$$

$$\varphi_{L_{\text{dec}}} = \frac{m \cdot a_{\text{dec}}}{N}$$

Slope costs:

$$S_L = \min \{ \text{slope}_{\text{long_uphill}}, \text{slope}_{\text{long_downhill}}, \text{slope}_{\text{side}} \}$$

$$\text{slope}_{\text{long_uphill}} = \min \{ \text{slope}_{lu_d}, \text{slope}_{lu_s} \}$$

$$\text{slope}_{lu_s} = \tan^{-1} \frac{l_2}{h}$$

$$\left\{ \begin{array}{l} F_t = F_f + F_i \\ F_t = \frac{T_{tq} \cdot i}{r} \\ F_f = \varphi \cdot G \cdot \cos(\text{slope}_{lu_d}) \\ F_i = G \cdot \sin(\text{slope}_{lu_d}) \end{array} \right.$$

Holonomic Traverability Optimization

4

Traversability Analysis and Path Planning for Autonomous Wheeled Vehicles on Rigid Terrains

Some of the parameters they take into consideration for their model

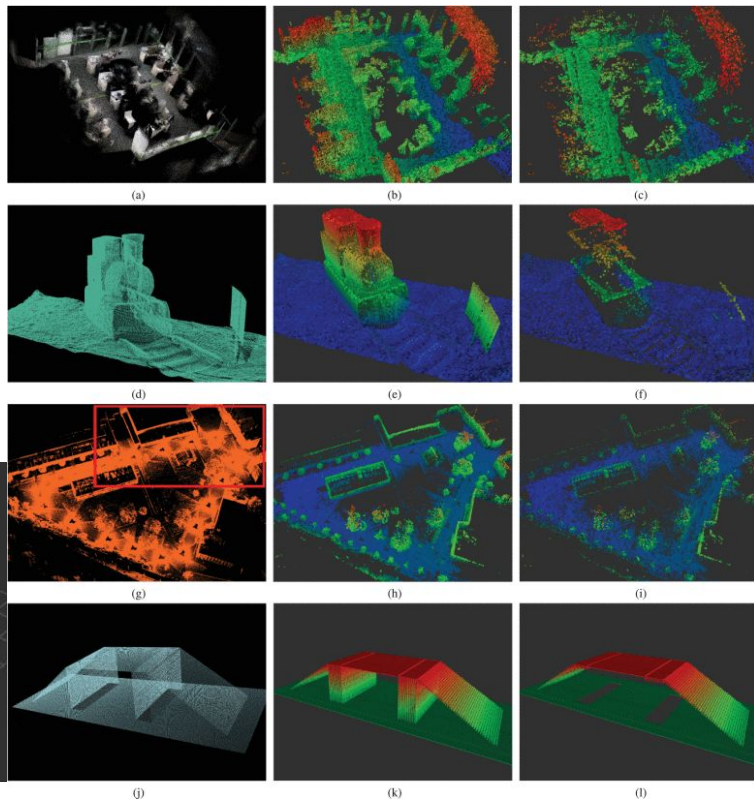
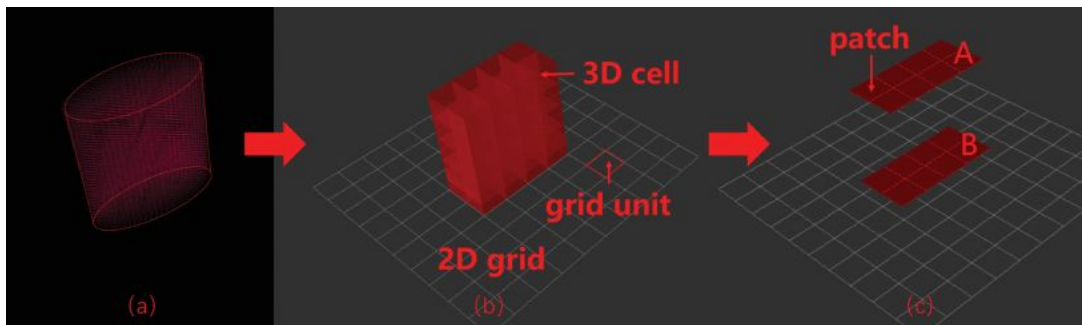
Parameter	Value	Units
Vehicle Length	4.84	m
Vehicle Width	2	m
Minimum Ground Clearance	0.3	m
Wheel Radius	0.42	m
Wheelbase	4	m
Maximum Engine Torque	300	N·m
Maximum Engine Power	110	kW
Maximum Braking Torque	4000	N·m
Minimum Turning Radius	5	m
Maximum Steering Angle	135	°

Holonomic Traverability Optimization

5

An Efficient Spatial Representation for Path
Planning of Ground Robots in 3D Environments
IEEE Journals & Magazine

2.5D Normal Distributions Transform Map
(NDT): 2.5D including **overhangs** and
bridges



Holonomic Traversability Optimization

6

3D traversability analysis and path planning based on mechanical effort for UGVs in forest environments - ScienceDirect

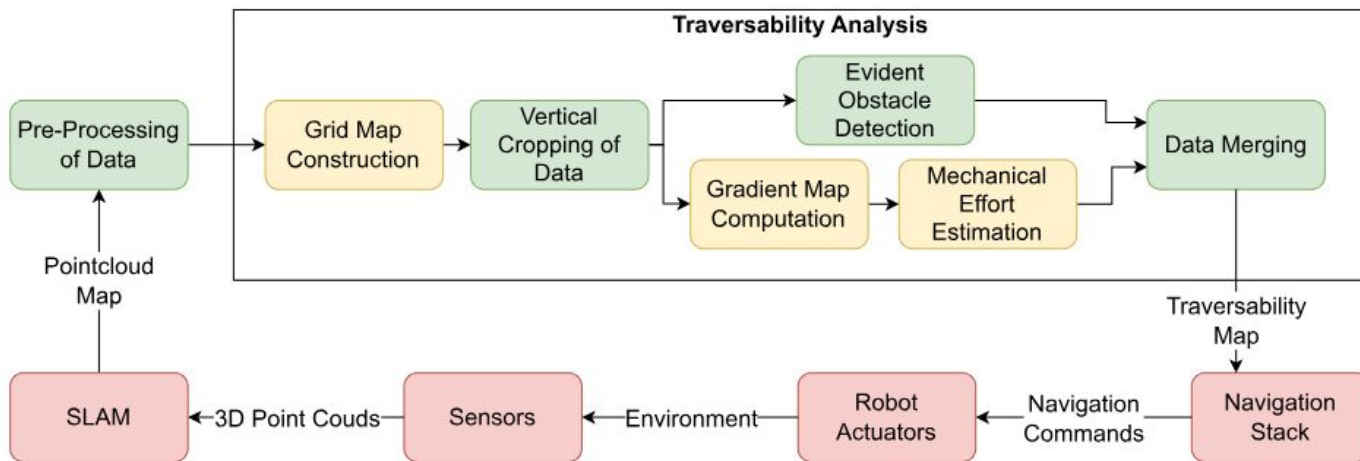


Fig. 1. Proposed System architecture. Green boxes: novel modules; Yellow boxes: modified modules; Red boxes: off-the-shelf modules used “as is”. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Holonomic Traversability Optimization

6

3D traversability analysis and path planning based on mechanical effort for UGVs in forest environments - ScienceDirect

Step	What Happens	Purpose
Pre-processing	Take the global 3D point cloud from SLAM, Remove points outside robot workspace (crop ROI) and noise/outliers, downsample with a 3D voxel grid filter	Reduce data size, clean up cloud, make processing faster
Grid Map Construction	Project all 3D points onto a 2D grid based on x,y location. Each cell stores a list of heights (z-values).	Organize data spatially
Vertical Cropping	Within each cell (list of z-values), take the lowest point (ground estimate), then remove only canopy, but keep points slightly above ground (tree trunks, rocks).	Approx ground surface
Gradient Estimation	Take the median height for gradient calculation to avoid noise. Use final height of each cell to compute slope between neighboring cells.	Measure slope ↔ energy cost
Evident Obstacle Detection	Use height range/variance in cell to decide if it's a rock/tree.	Mark obstacles
Mechanical Effort	Combine slope + direction of motion.	Energy cost
Data Merge	Combine slope + obstacles → costmap	Final traversability

Holonomic Traversability Optimization

6

3D traversability analysis and path planning based on mechanical effort for UGVs in forest environments - ScienceDirect

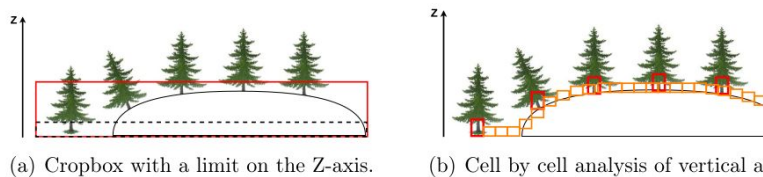


Fig. 2. Cropping heuristics. The one illustrated in (b) was ultimately selected.

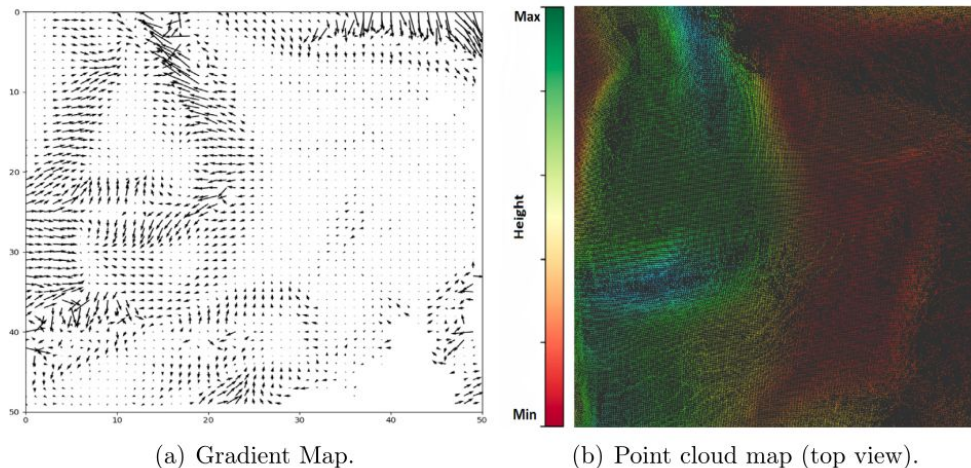
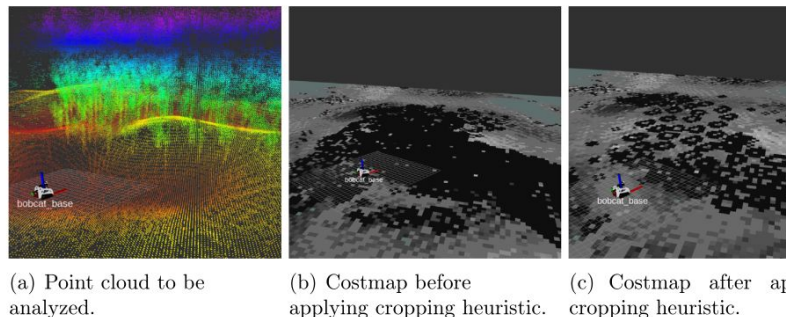


Fig. 4. Output representation of the gradient analysis stage.

Holonomic Traversability Optimization

6

[3D traversability analysis and path planning based on mechanical effort for UGVs in forest environments - ScienceDirect](#)

Gradient computation

$$\frac{\partial f}{\partial \lambda} = \frac{f(\lambda + h) - f(\lambda - h)}{2h},$$

$$\frac{\partial f}{\partial x} \approx \frac{f(i + 1, j) - f(i - 1, j)}{2h}$$

$$\frac{\partial f}{\partial y} \approx \frac{f(i, j + 1) - f(i, j - 1)}{2h}$$

$$\nabla f(x_m, y_n) = \begin{cases} \nabla f(x_m, y_n), & \|\nabla f(x_m, y_n)\| < \nabla_{th} \\ \nabla_{th}, & \text{otherwise,} \end{cases}$$

Evident Obstacle Detection

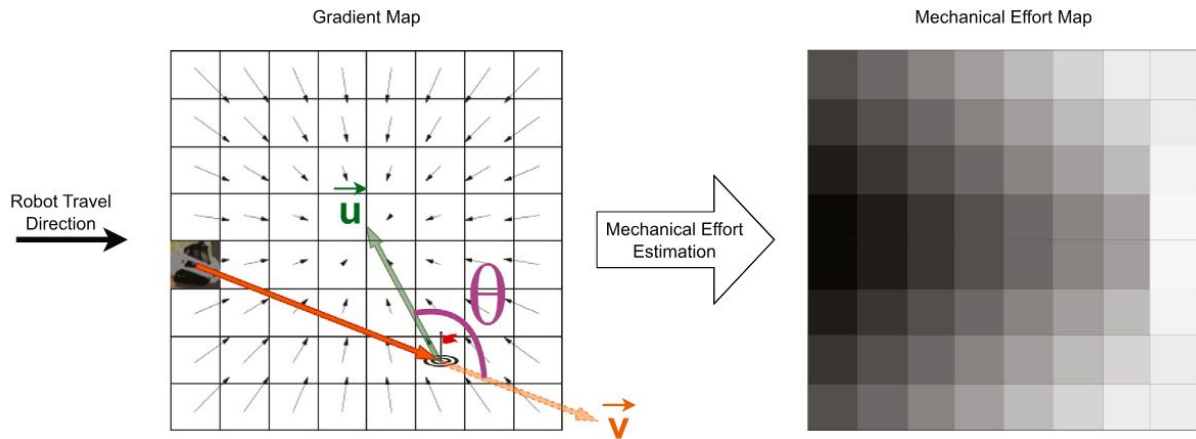
$$obstacle = \begin{cases} \text{true,} & \begin{cases} \mu_{(x,y)} > \mu_{th} \\ \sigma_{(x,y)}^2 > \sigma_{th}^2 \\ (max_{(x,y)} - min_{(x,y)}) > \gamma \end{cases} \\ \text{false,} & \text{otherwise,} \end{cases}$$

Holonomic Traversability Optimization

6

3D traversability analysis and path planning based on mechanical effort for UGVs in forest environments - ScienceDirect

Mechanical Effort Estimation



Relies on a **locally updated effort** metric but lacks **global terrain reasoning** — it does not incorporate detailed terrain features such as **surface roughness** or **anisotropic slope effects**, and it evaluates mechanical effort **only at the current node**, without accumulating effort along the full path. May choose locally minimal effort moves that lead to globally inefficient or risky trajectories.

$$\gamma = \|\nabla f(x, y)\| \cdot \cos \theta,$$

6

[3D traversability analysis and path planning based on mechanical effort for UGVs in forest environments - ScienceDirect](#)

Data merging - Traversability Cost Map

0: steep descent (requires the less energy)

50: flat terrain

99: steep ascent

100: untraversable ascent/descent/roughness or obstacle

Holonomic Traversability Optimization

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3D traversability analysis and path planning based on mechanical effort for UGVs in forest environments - ScienceDirect

Rest of the Pipeline (red)

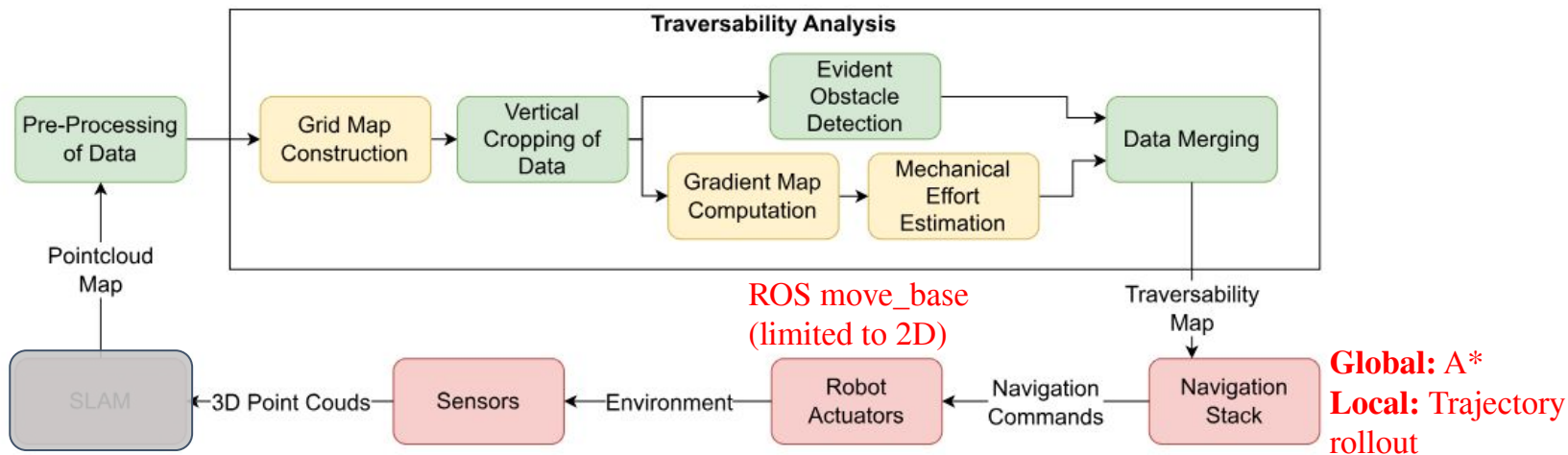
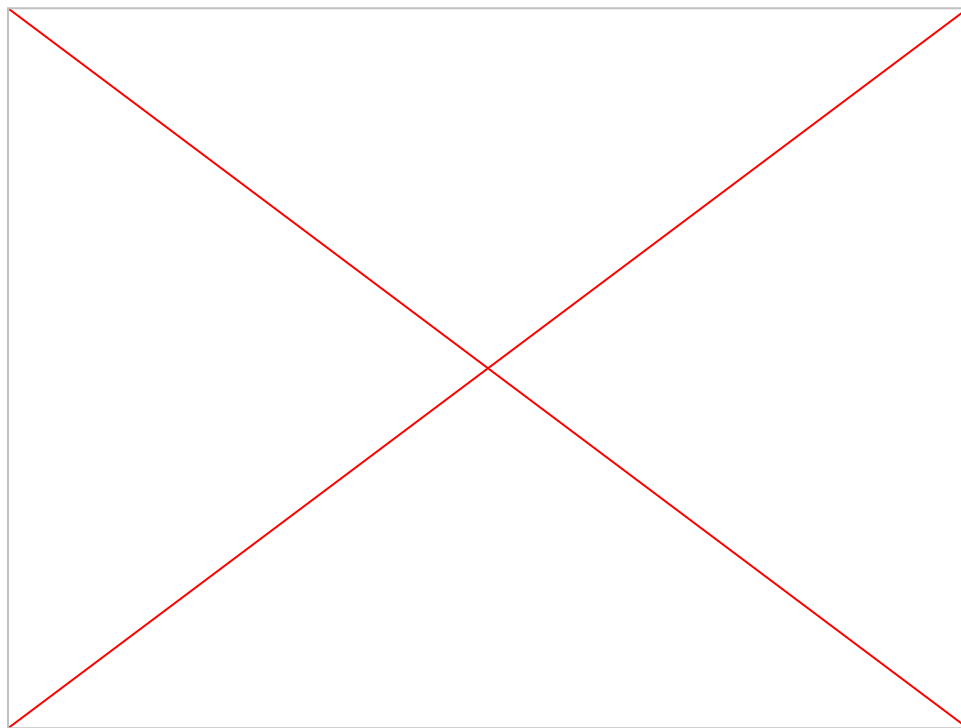


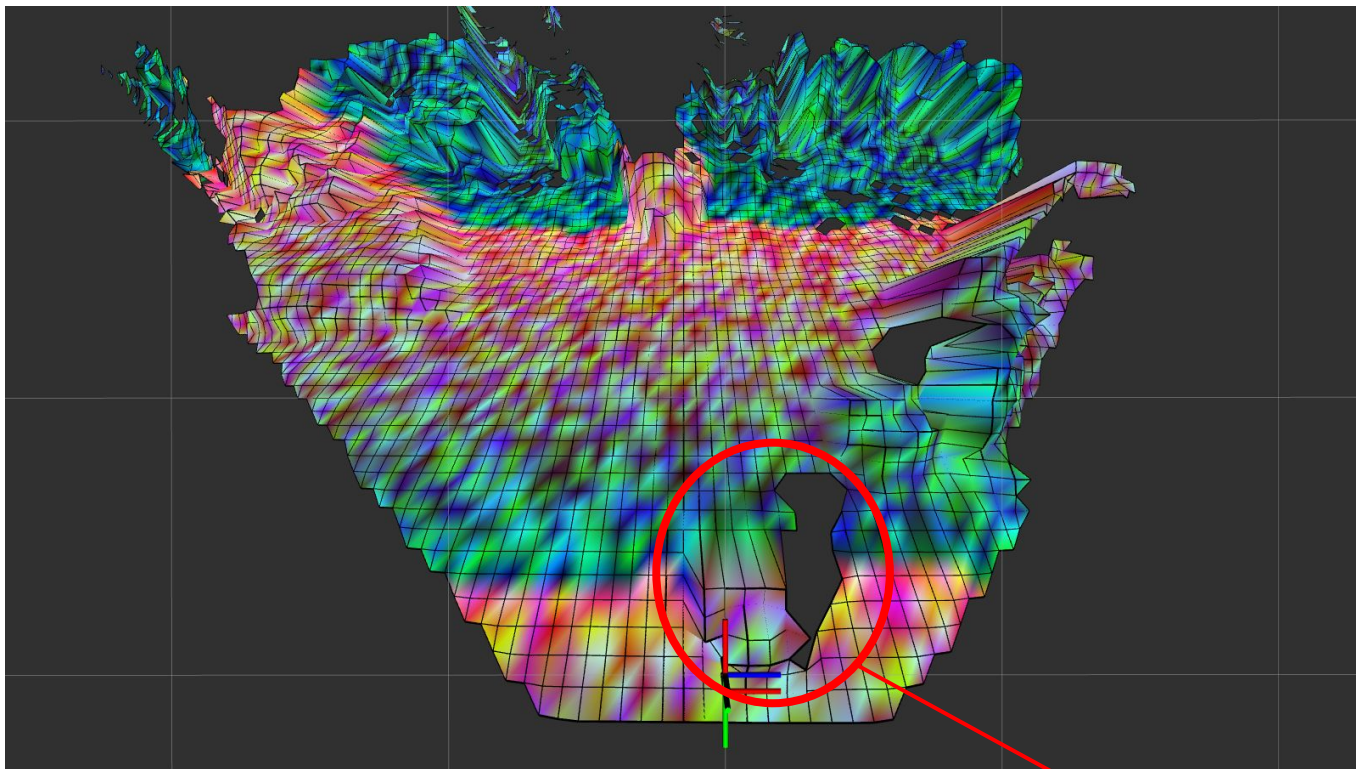
Fig. 1. Proposed System architecture. Green boxes: novel modules; Yellow boxes: modified modules; Red boxes: off-the-shelf modules used “as is”. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

CV: Updates



PCL Visualization - next steps: 2.5D map construction

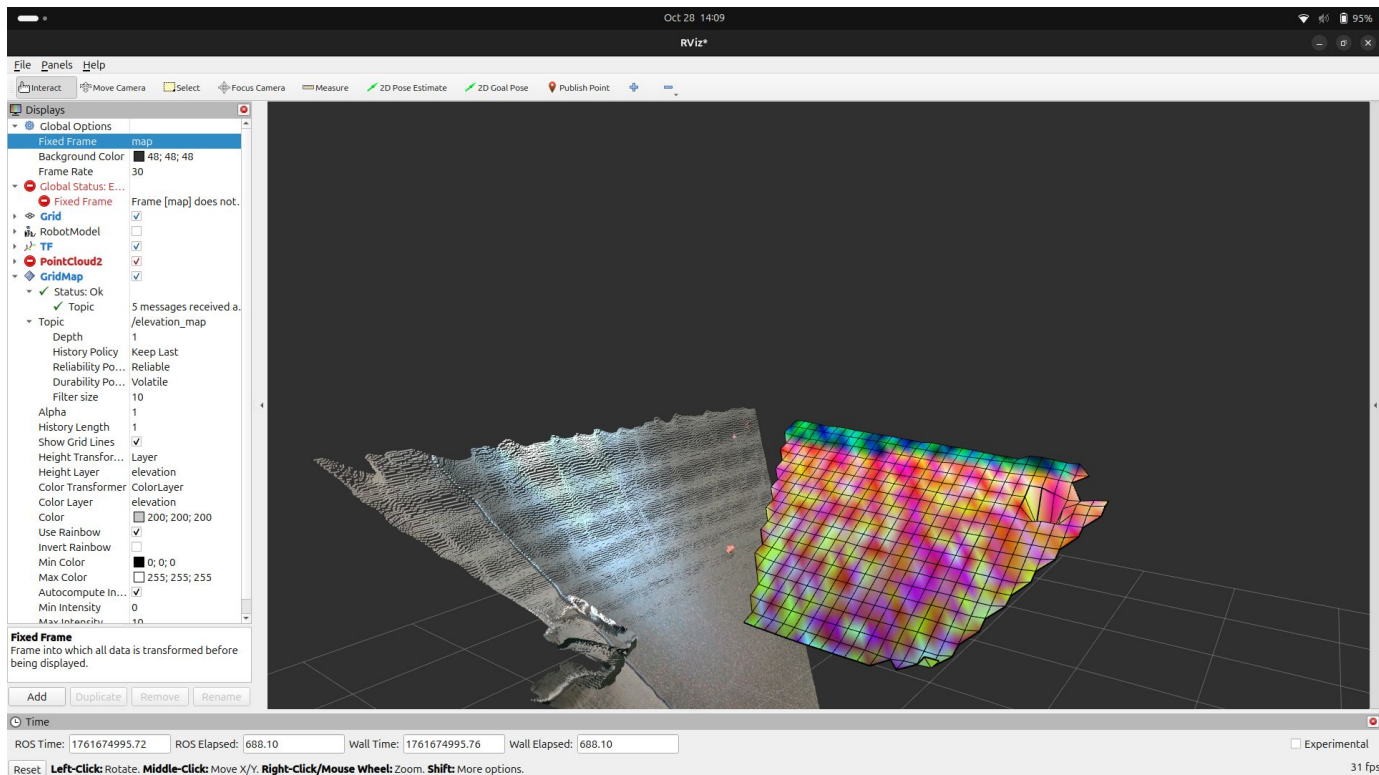
CV: Updates



2.5D map construction from PCL (Local)

Box

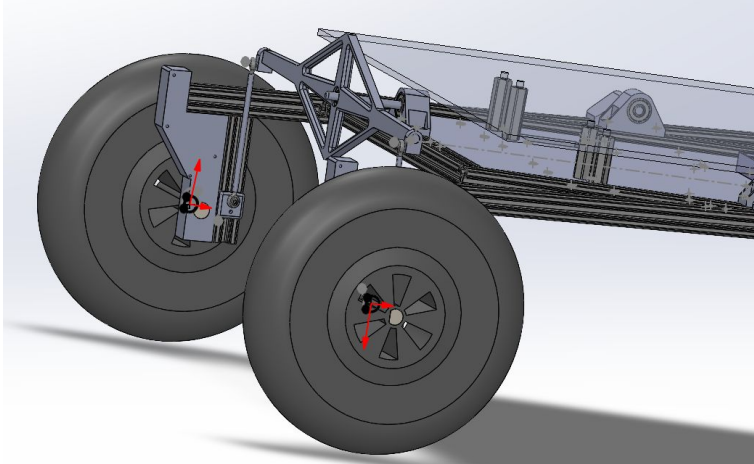
CV: Updates



Mismatch between PCL and gridmap frames to be fixed

Hardware: Updates

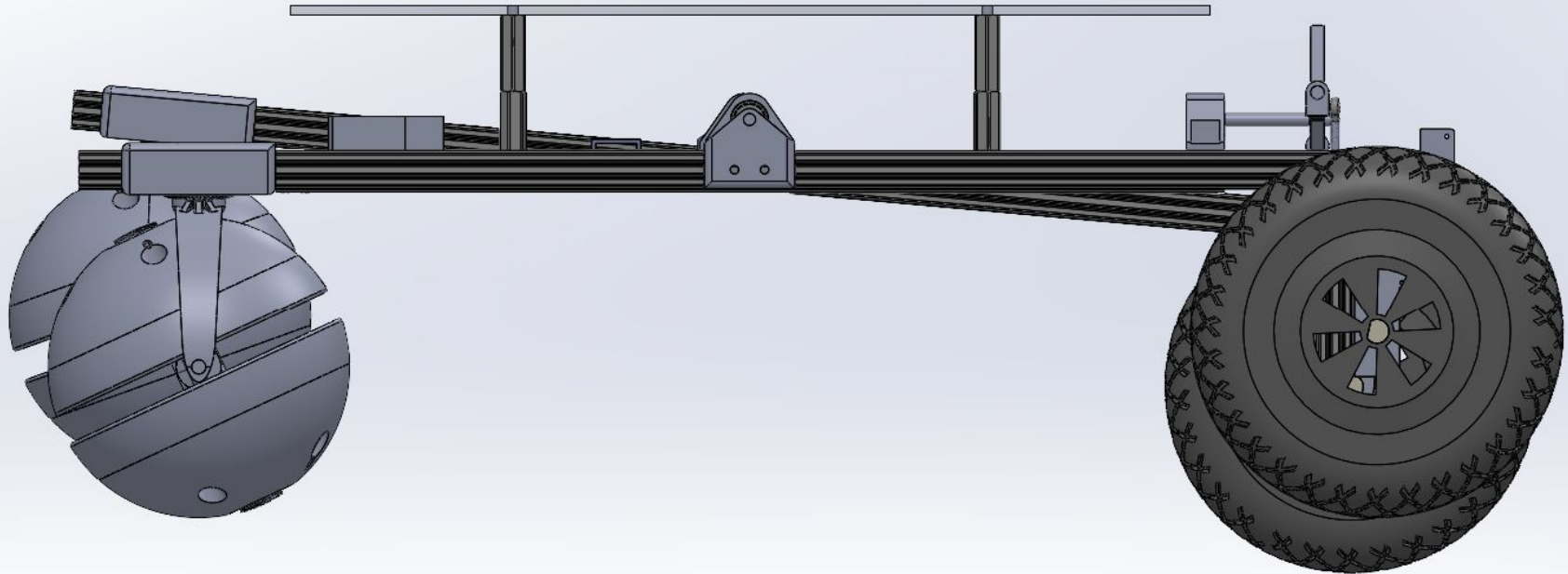
- Gimbal assembled
- Designing electronics layout
- Begin manufacturing
- Finished base design



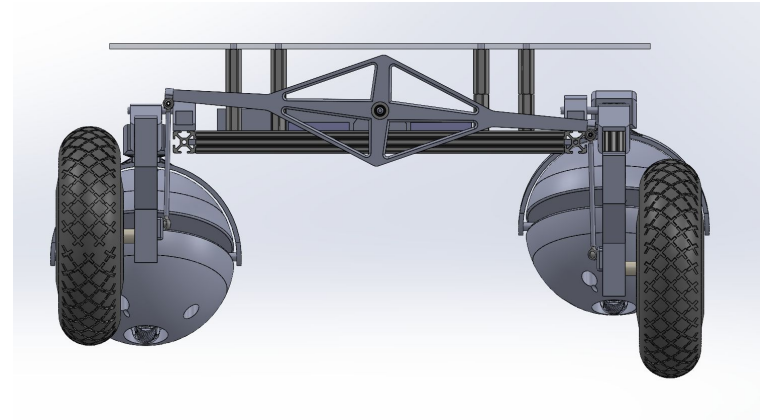
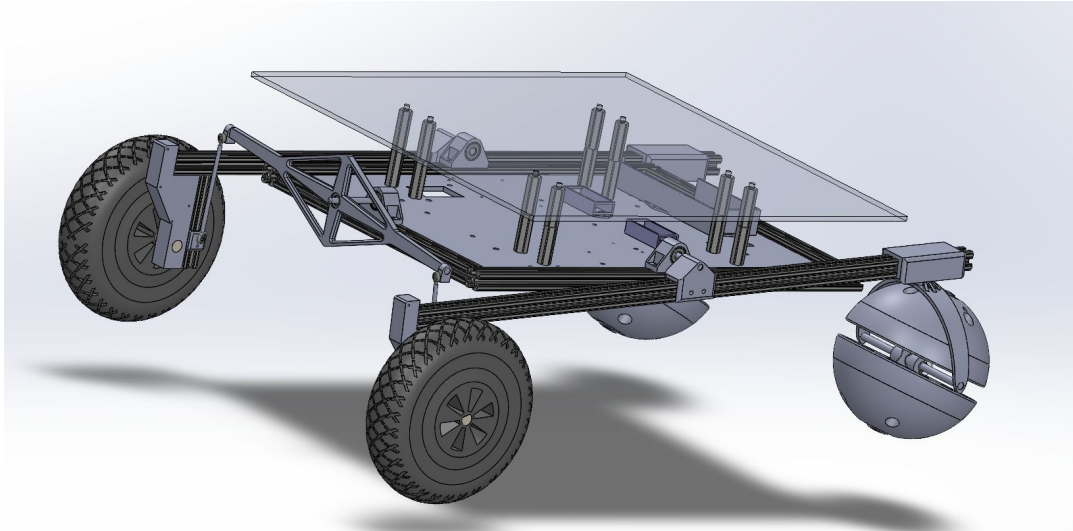
Assembled Gimbal Video



Spherical Caster Update

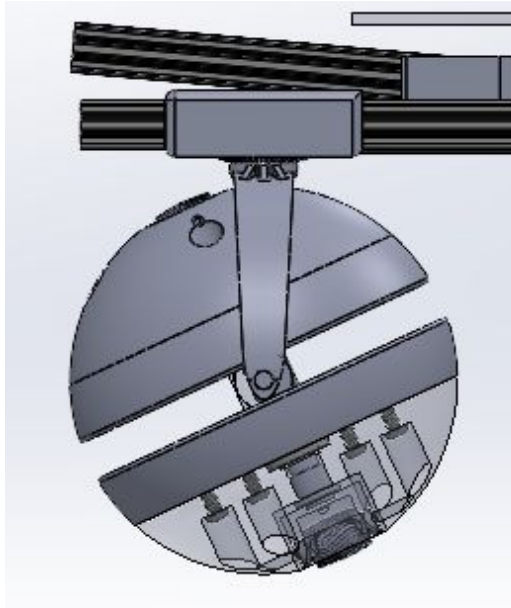


Full CAD

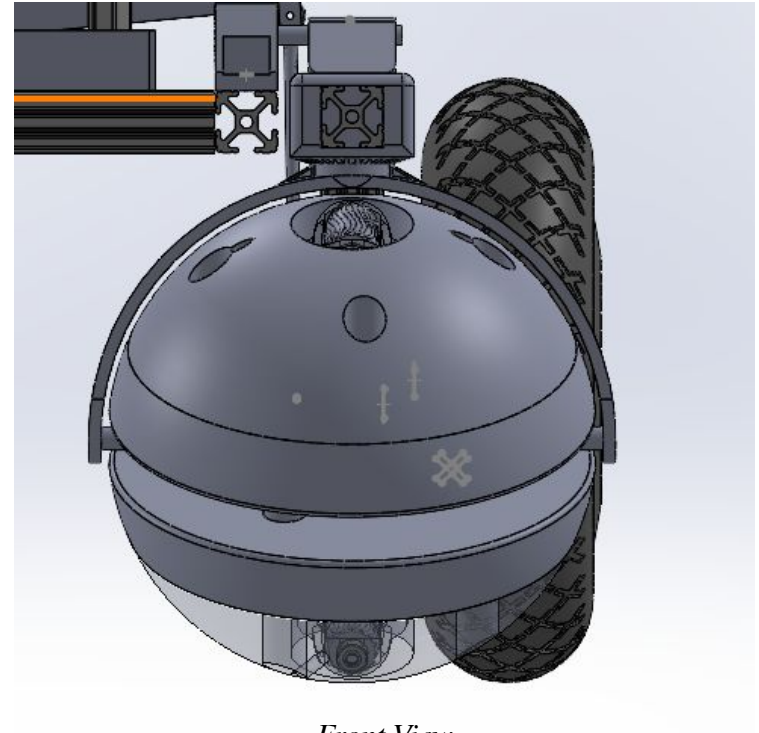


Spherical Caster Update

We will try different angles of the
caster fork. It is vertical for now

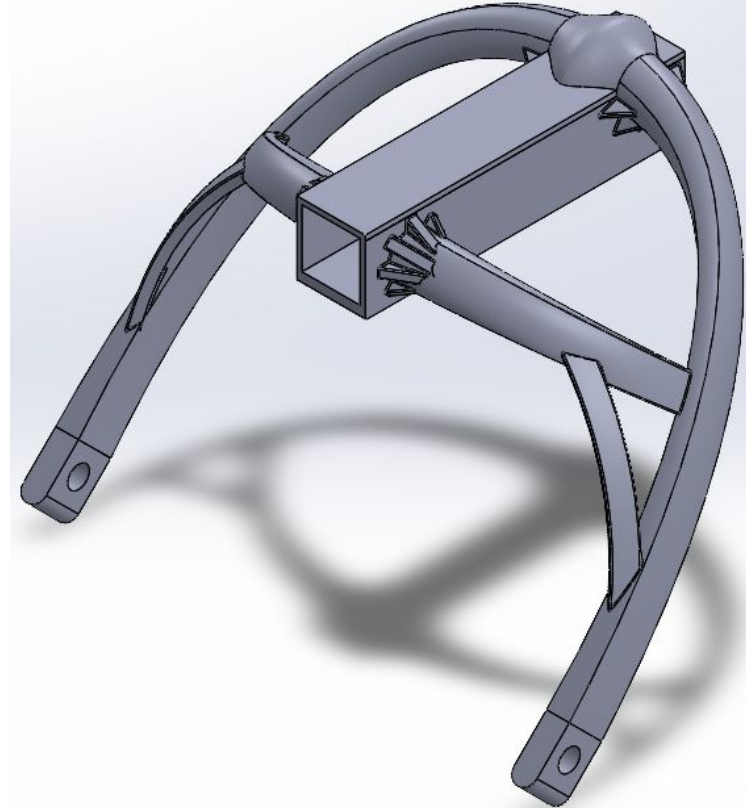
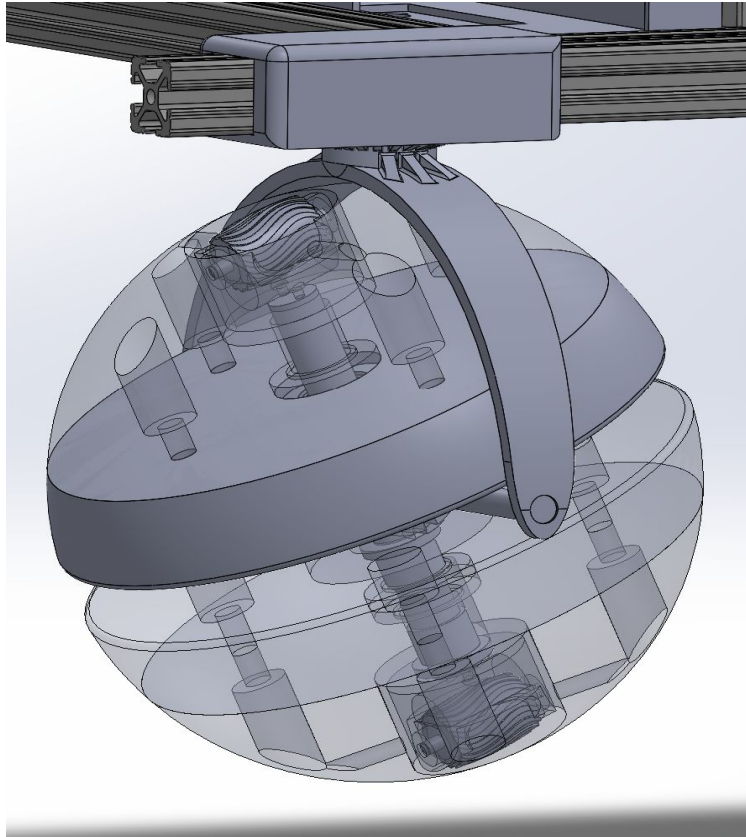


Side View

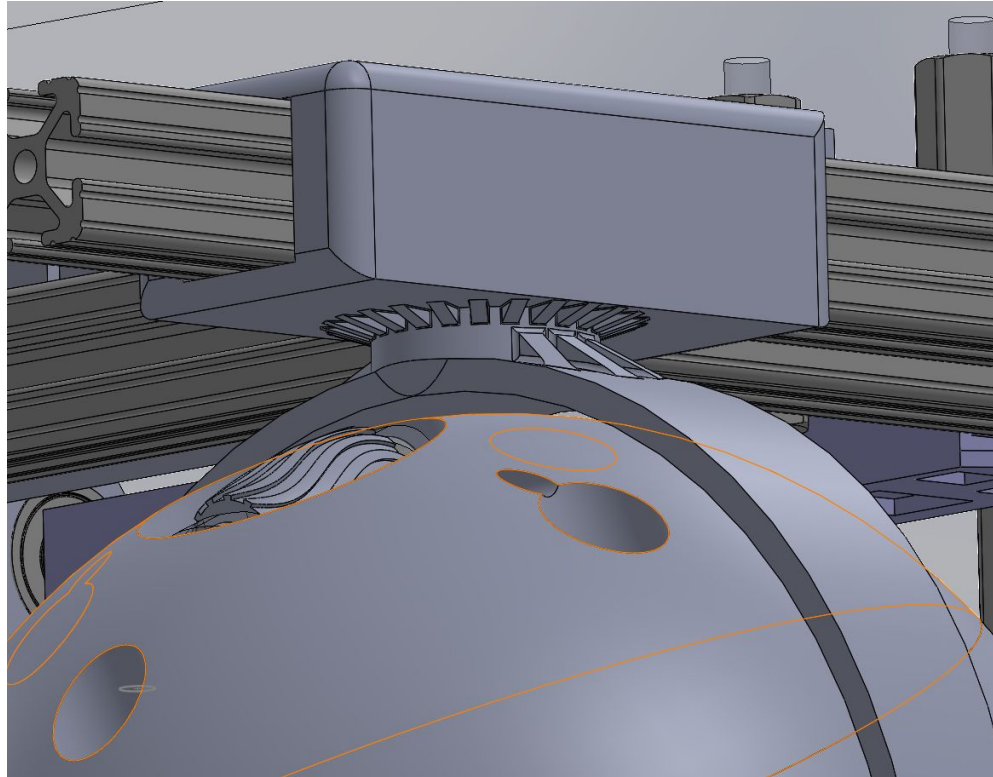


Front View

Spherical Caster Update



Spherical Caster Update



Hardware: Review with Giuseppe

Caster:

- Change shape of nuts (facilitate assembling)
- Gap between hemispheres,
- Bigger roller
- Bigger fork
- Rocks might be getting inside the wheel
- 8cm clearance because of fork might really be too low
- Try regular, and try at different angles (0° , 15° , 30° , 45°)

Base:

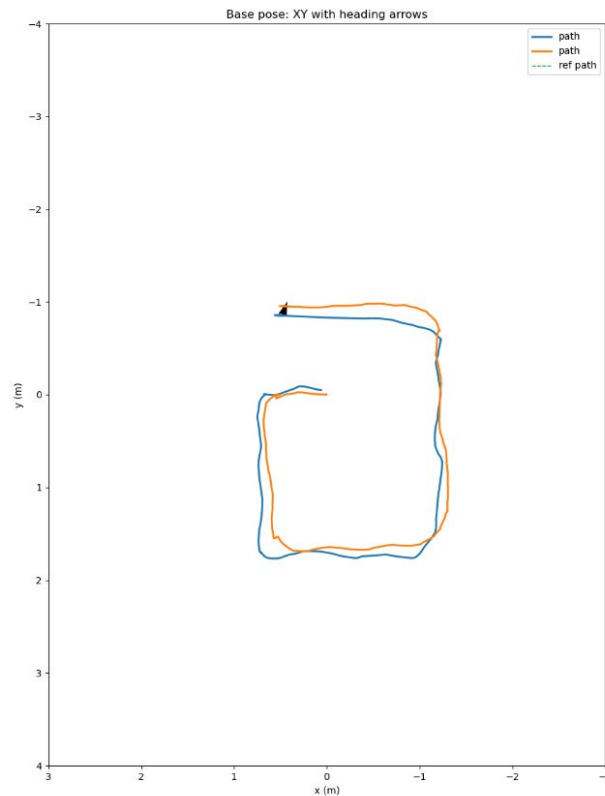
- Rails too thin
- Move Leverage link closer to base

Build Process + Other Updates

- Modified electronic layout
- Increased surface height to increase rocker mobility
- Gathering/planning materials
- Build process procedure

Odometry: Updates

- Imu EKF fused

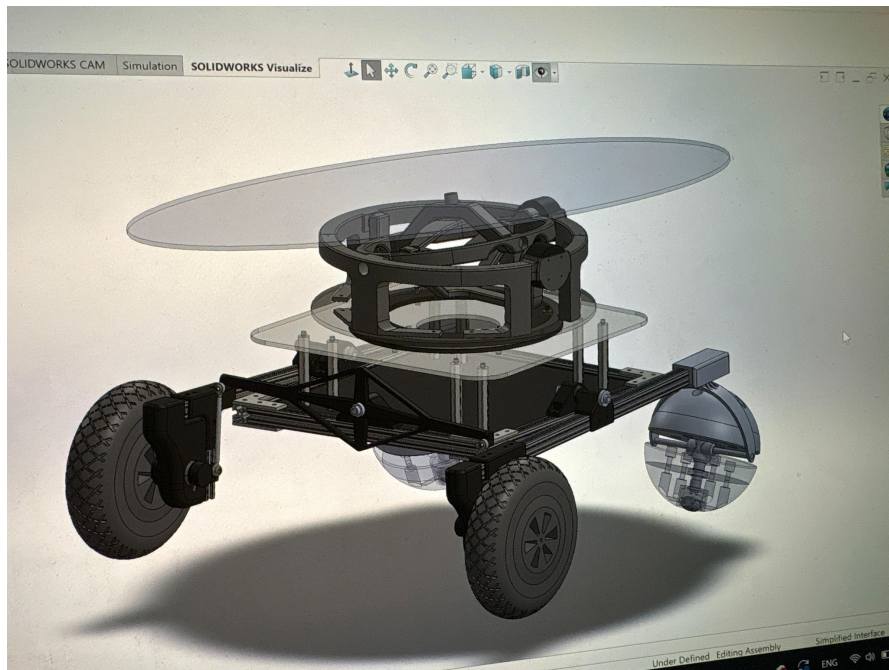




(11/06)

Weekly Updates

Hardware: CAD Update



Updated CAD



(11/14)

Weekly Updates

Traversability Optimization Papers

GMU: Estimating Traversability for Wheeled Mobility on Vertically Challenging Terrain

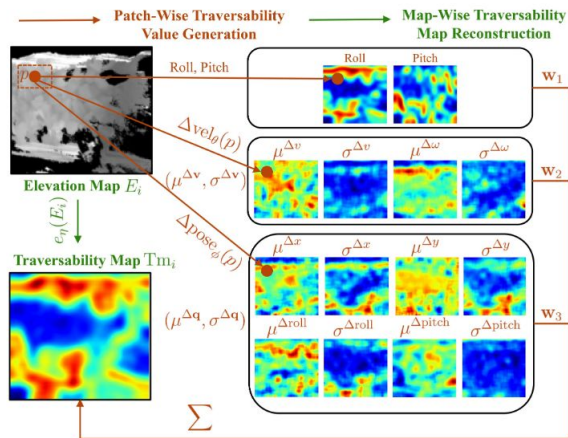


Fig. 2: **TNT Overview.** Based on terrain patches p on the elevation map E_i , three predictors produce roll and pitch angles, velocity discrepancy $(\mu^{\Delta v}, \sigma^{\Delta v})$, and pose prediction discrepancy $(\mu^{\Delta q}, \sigma^{\Delta q})$, which are combined by w_1 , w_2 , and w_3 to generate patch-wise traversability values; All traversability values form a traversability map Tm_i , which a map-wise traversability map estimator ($e_\eta(\cdot)$) learns to reconstruct based on terrain patch E_i .

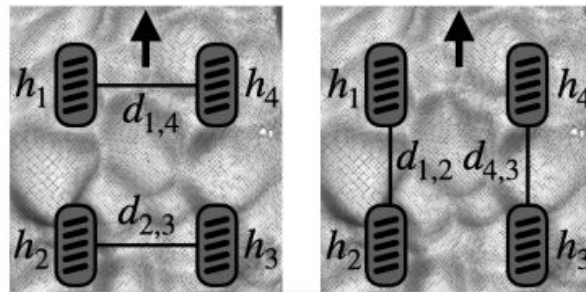


Fig. 3: Roll and Pitch Model.

$$\text{roll, pitch} = \frac{1}{2} \left\| \sum_{(i,j) \in \mathcal{I}_{\text{roll, pitch}}} \left(\arctan \left(\frac{h_i - h_j}{d_{i,j}} \right) \right) \right\|, \quad (1)$$

Traversability Optimization Papers

GMU: [Estimating Traversability for Wheeled Mobility on Vertically Challenging Terrain](#)

Patch-Wise Traversability Value Generation:

1. Roll and Pitch Angles

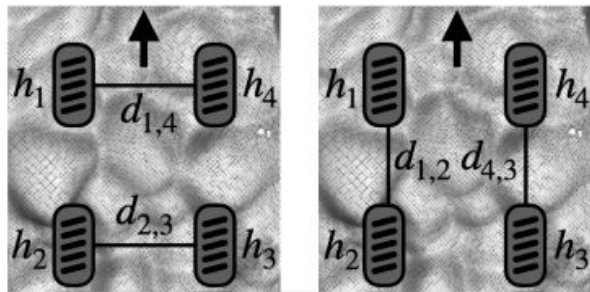


Fig. 3: Roll and Pitch Model.

patch based on a quasi-static physics-based model. The roll and pitch angles are estimated based on the elevation value h_i of the terrain patch pixel under each vehicle wheel i using trigonometry:

$$\text{roll, pitch} = \frac{1}{2} \left\| \sum_{(i,j) \in \mathcal{I}_{\text{roll, pitch}}} \left(\arctan \left(\frac{h_i - h_j}{d_{i,j}} \right) \right) \right\|, \quad (1)$$

where $d_{i,j}$ is the distance between the wheels i, j , and the subscript sets for roll and pitch are $\mathcal{I}_{\text{roll}} = \{(1, 4), (2, 3)\}$ (Fig. 3 left) and $\mathcal{I}_{\text{pitch}} = \{(1, 2), (4, 3)\}$ (Fig. 3 right), respectively. Notice that Eqn. (1) only considers the terrain patch, not vehicle state and action, and the physics model is deterministic.

Traversability Optimization Papers

GMU: [Estimating Traversability for Wheeled Mobility on Vertically Challenging Terrain](#)

Patch-Wise Traversability Value Generation:

2. Velocity Discrepancy

$$\Delta \mathbf{v} = \mathbf{v}_{\text{commanded}} - \mathbf{v}_{\text{actual}},$$

$$(\mu^{\Delta \mathbf{v}}, \sigma^{\Delta \mathbf{v}}) = \Delta \text{vel}_{\theta}(p), \quad (2)$$

where $\mu^{\Delta \mathbf{v}}, \sigma^{\Delta \mathbf{v}}$ are the mean and standard deviation of the difference in linear (Δv) and angular ($\Delta \omega$) velocity, whereas the estimator $\Delta \text{vel}_{\theta}(\cdot)$ is parameterized by learnable parameters θ .

Given a dataset of ground truth $\mathcal{D}_{\Delta \mathbf{v}} = \{\Delta \mathbf{v}_i, p_i\}_{i=1}^N = \{(\Delta v_i, \Delta \omega_i), p_i\}_{i=1}^N$ computed by the difference between the outputs of a vehicle state estimator (e.g., internal Visual-Inertial Odometry or external GPS-RTK or motion capture system) and velocity commands, we represent $\Delta \text{vel}_{\theta}(\cdot)$ as a learnable neural network and learn the parameters θ using a negative log likelihood loss:

$$\mathcal{L}_{\mathcal{D}_{\Delta \mathbf{v}}} = \sum_{\substack{\{\Delta \mathbf{v}_i, p_i\} \\ \in \mathcal{D}_{\Delta \mathbf{v}}}} \frac{1}{2} \left(\log(\sigma_i^{\Delta \mathbf{v}^2}(p_i)) + \frac{(\mu_i^{\Delta \mathbf{v}}(p_i) - \Delta \mathbf{v}_i)^2}{\sigma_i^{\Delta \mathbf{v}^2}(p_i)} \right).$$

Traversability Optimization Papers

GMU: [Estimating Traversability for Wheeled Mobility on Vertically Challenging Terrain](#)

Patch-Wise Traversability Value Generation:

3. Pose Prediction discrepancy

another data-driven estimator:

$$(\mu^{\Delta \mathbf{q}}, \sigma^{\Delta \mathbf{q}}) = \Delta \text{pose}_{\phi}(p), \quad (4)$$

parameterized by learnable parameters ϕ . Another negative log likelihood loss, similar to Eqn. (3) but with the $\Delta \mathbf{v}$ components replaced by $\Delta \mathbf{q}$, is used to learn ϕ in Eqn. (4):

$$\mathcal{L}_{\mathcal{D}_{\Delta \mathbf{q}}} = \sum_{\substack{\{\Delta \mathbf{q}_i, p_i\} \\ \in \mathcal{D}_{\Delta \mathbf{q}}}} \frac{1}{2} \left(\log(\sigma_i^{\Delta \mathbf{q}^2}(p_i)) + \frac{(\mu_i^{\Delta \mathbf{q}}(p_i) - \Delta \mathbf{q}_i)^2}{\sigma_i^{\Delta \mathbf{q}^2}(p_i)} \right).$$

GMU: [Estimating Traversability for Wheeled Mobility on Vertically Challenging Terrain](#)

Patch-Wise Traversability Value Generation:

Resulting Traversability eqn for each patch p :

$$\text{Traversability}(p) = \mathbf{w}_1 \cdot [\text{roll}, \text{pitch}]^T + \mathbf{w}_2 \cdot [\mu^{\Delta \mathbf{v}}, \sigma^{\Delta \mathbf{v}}]^T + \mathbf{w}_3 \cdot [\mu^{\Delta \mathbf{q}}, \sigma^{\Delta \mathbf{q}}]^T, \quad (5)$$

Traversability Optimization Papers

An Efficient Trajectory Planner for Car-like Robots on Uneven Terrain

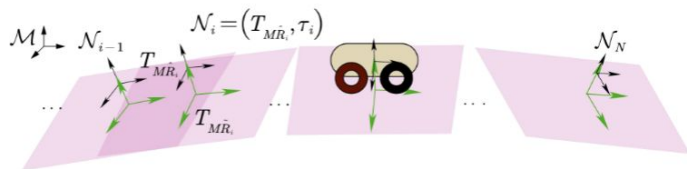


Fig. 3: Representation of trajectory for motion planning in complex terrain. The planner trajectory is defined as a series of nodes extended by the sampling tree. Each node $\mathcal{N}_i$ consists of a 6D pose $T_{M\hat{R}_i}$ and a traversability τ_i . Based on the set of selected terrain surface points Ω_i , the SVD method is used to fit a plane P_i . Keeping the distance between the nodes satisfies kinematic vehicle constraints.

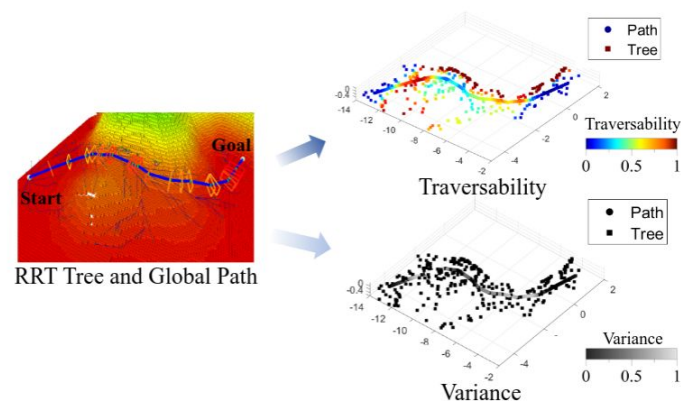
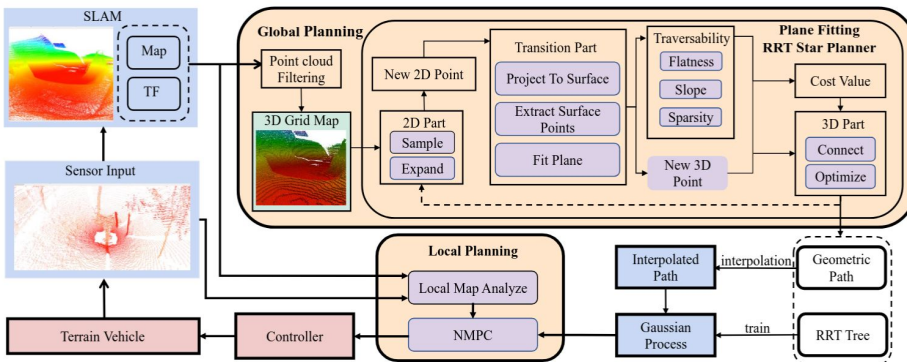


Fig. 5: Traversability estimation of interpolated path points using GPR. The global path and the RRT tree generated by the PF-RRT* algorithm on uneven terrain are shown on the left side. The traversability and variance of global path interpolation points estimated by GPR prediction are shown on the right side.

Traversability Optimization Papers

An Efficient Trajectory Planner for Car-like Robots on Uneven Terrain

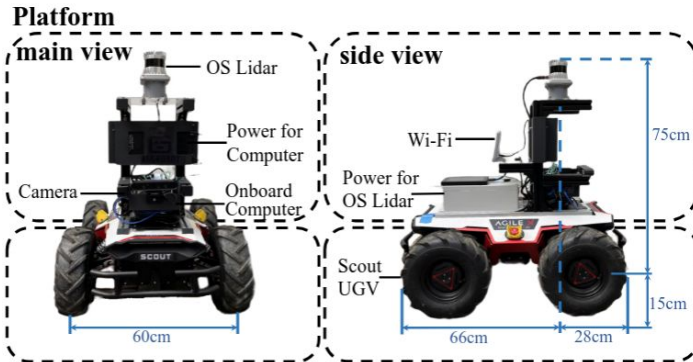


Fig. 7: Our PUTN vehicle platform for the experiment, equipped with OS0 lidar and a camera(only for front view in the video). Two batteries are installed to power the lidar and the onboard computer. Wi-Fi is used for communication.

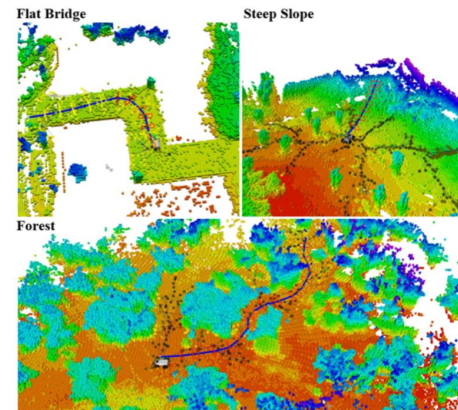


Fig. 9: Experiments in flat bridge, steep slope, and forest. The real scenarios corresponding to these experiments can be found in Fig.1. The video of real-world experiments is available at <https://www.youtube.com/watch?v=3ZK-Ut29hLI>.

Traversability Optimization Papers

An Efficient Trajectory Planner for Car-like Robots on Uneven Terrain

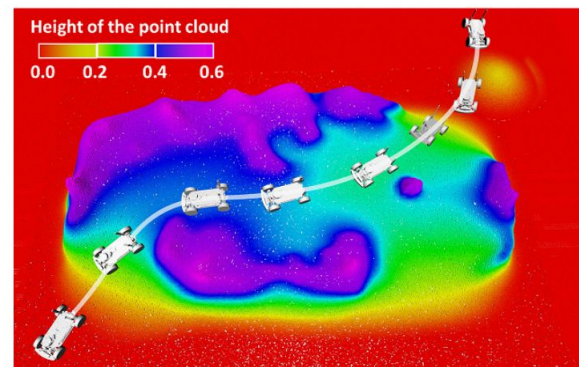
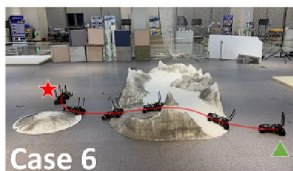
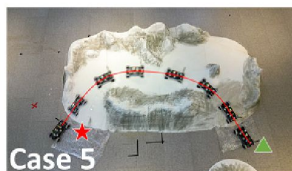
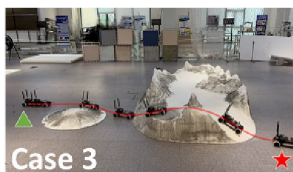
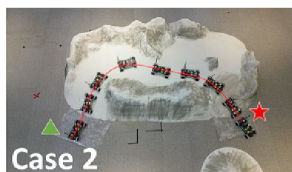
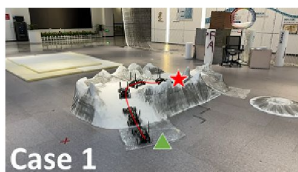
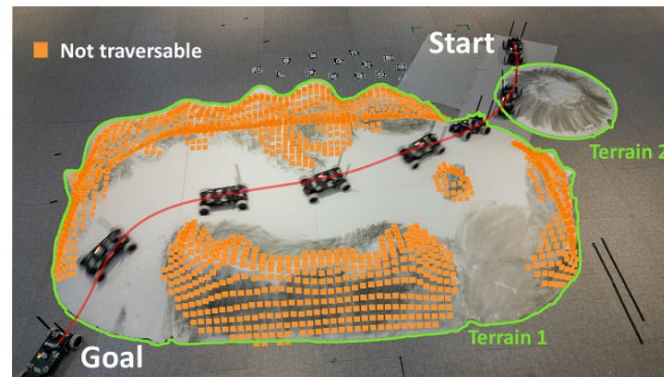
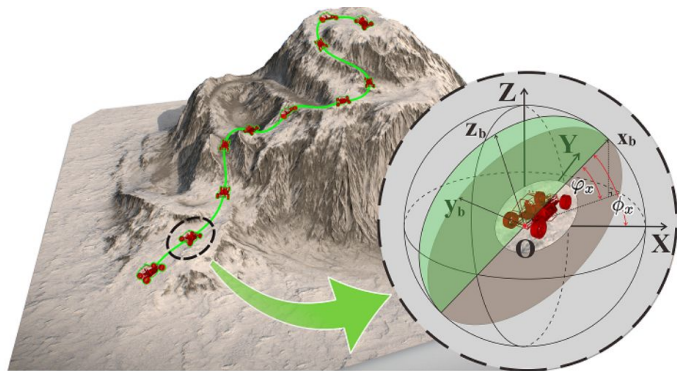
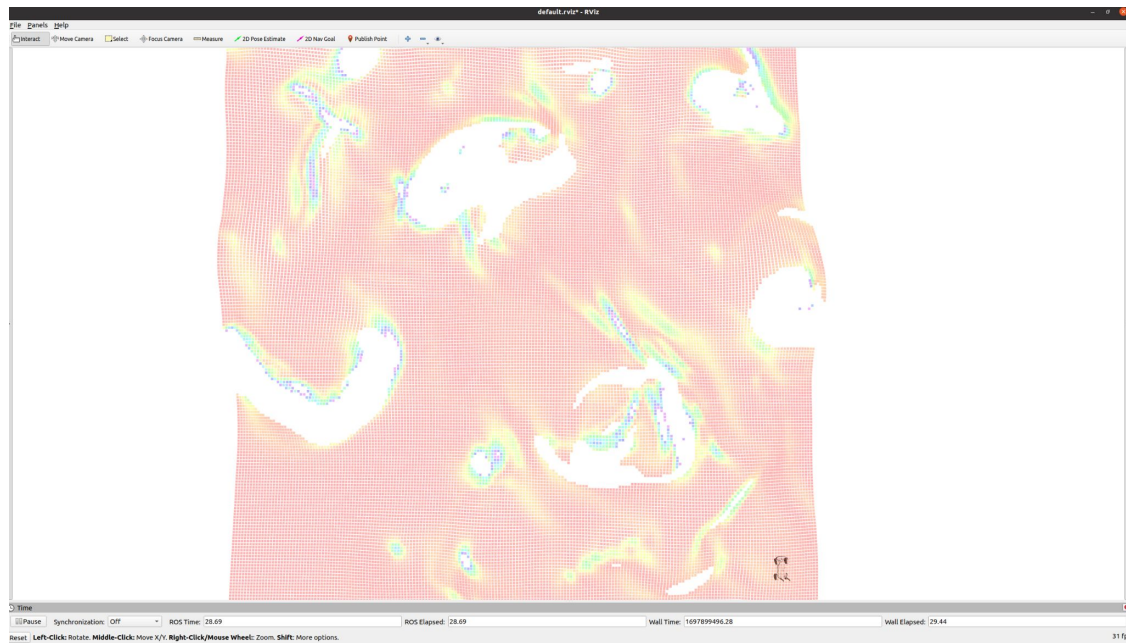


Fig. 6: The motion diagram of car-like robot in the real-world experiments, where the red curves are the planned trajectories. The green triangles indicate the starting points and the red pentagrams indicate the end points.

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[An Efficient Trajectory Planner for Car-like Robots on Uneven Terrain](#)

ROS1..



Github Contains .pcd examples

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An Efficient Trajectory Planner for Car-like Robots on Uneven Terrain

Optimal Control Problem (NMPC) - Non-Holonomic

$$\min_{\mathbf{u}(t)} \int_0^{t_f} \tau(\mathbf{s}(t), \mathbf{s}^{(1)}(t), \dots, \mathbf{s}^{(s)}(t), \mathbf{u}(t)) dt + \rho(t_f) \quad (1)$$

$$s.t. \quad \dot{\mathbf{s}}(t) = f(\mathbf{s}(t), \mathbf{u}(t)), \quad (2)$$

$$\text{Ter}(\mathbf{s}(t)) = 0, \quad (3)$$

$$C_p(\mathbf{s}(t), \mathbf{s}^{(1)}(t), \dots, \mathbf{s}^{(s)}(t), \mathbf{u}(t)) \preceq \mathbf{0}, \quad (4)$$

$$\min_{\mathbf{c}_{xy}, \mathbf{c}_\theta, \mathbf{T}_{xy}, \mathbf{T}_\theta} \int_0^{T_s} (\mathbf{j}(t)^T \mathbf{j}(t) + \rho_{\text{ter}} \sigma(\mathbf{x}(t))) dt + \rho_T T_s \quad (28)$$

$$s.t. \quad \dot{x} \sin \theta - \dot{y} \cos \theta = 0, \quad (29)$$

$$\mathbf{M}_{xy} \mathbf{c}_{xy} = \mathbf{b}_{xy}, \quad \mathbf{M}_\theta \mathbf{c}_\theta = \mathbf{b}_\theta, \quad (30)$$

$$\mathbf{T}_{xy} \succeq \mathbf{0}, \quad \mathbf{T}_\theta \succeq \mathbf{0}, \quad (31)$$

$$v_x^2 - v_{max}^2 \leq 0, \quad (32)$$

$$a_x^2 - a_{mlon}^2 \leq 0, \quad (33)$$

$$a_y^2 - a_{mlat}^2 \leq 0, \quad (34)$$

$$\frac{\omega_z^2}{v_x^2 + \delta_+} - \frac{\tan^2 \delta_{max}}{L_w^2} \leq 0, \quad (35)$$

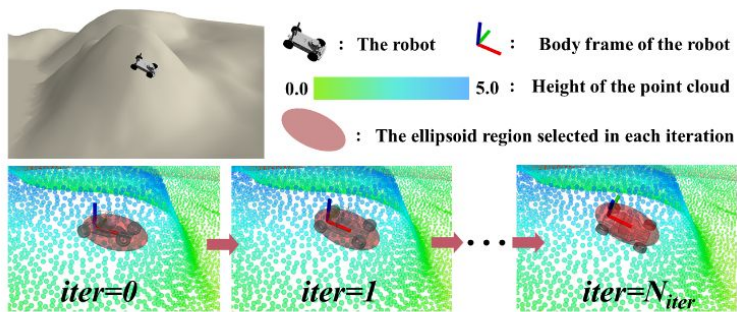
$$c_{min} - \cos \xi \leq 0, \quad (36)$$

$$\sigma(\mathbf{x}) - \sigma_{max} \leq 0, \quad (37)$$

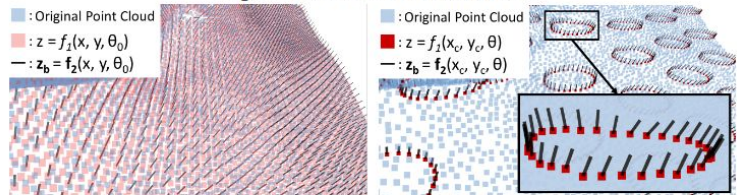
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Terrain Pose Mapping



(a) process of the construction.



In this work, we consider the terrain contact constraint (4) brings about actually a mapping relation called *terrain pose mapping* $\mathcal{F} : SE(2) \mapsto \mathbb{R} \times \mathbb{S}_+^2$, where $\mathbb{S}_+^2 \triangleq \{\mathbf{x} \in \mathbb{R}^3 \mid \|\mathbf{x}\|_2 = 1, \mathbf{x} \cdot \mathbf{b}_3 > 0\}$, $\mathbf{b}_3 = [0, 0, 1]^T$. This means that the state in $SE(2)$, which is originally on flat ground, is given height and attitude, with elements in $\mathbb{R}$ representing the height z , elements in $\mathbb{S}_+^2$ representing the body axis $\mathbf{z}_b$.

Let $\mathbf{x}_{yaw} = [\cos \theta, \sin \theta, 0]^T$ denote the direction of yaw angle, the mapping $\mathcal{F}$ can be expressed as two functions:

$$z = f_1(x, y, \theta), \quad (7)$$

$$\mathbf{z}_b = \mathbf{f}_2(x, y, \theta). \quad (8)$$

$$\mathbf{p} = [x, y, z]^T = [x, y, f_1(x, y, \theta)]^T, \quad (9)$$

$$\mathbf{y}_b = \frac{\mathbf{f}_2(x, y, \theta) \times \mathbf{x}_{yaw}}{\|\mathbf{f}_2(x, y, \theta) \times \mathbf{x}_{yaw}\|}, \quad (10)$$

$$\mathbf{x}_b = \mathbf{y}_b \times \mathbf{f}_2(x, y, \theta). \quad (11)$$

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Trajectory Parameterization (quintic piecewise polynomials - “mapping F allows the state space of the robot to still be SE(2))

$$\begin{aligned}x_i(t) &= c_{x_i}^T \gamma(t), \quad t \in [0, T_i]; \\y_j(t) &= c_{y_j}^T \gamma(t), \quad t \in [0, T_j]; \\\theta_k(t) &= c_{\theta_k}^T \gamma(t), \quad t \in [0, T_k];\end{aligned}$$

Piecewise Trajectories

$$\begin{aligned}\mathbf{x}(t) &= [x(t), y(t), \theta(t)]^T, \\ \mathbf{y}_{yaw} &= [-\sin \theta, \cos \theta, 0]^T,\end{aligned}$$

Entire Trajectory

$$\begin{aligned}v_x &= \frac{v}{\cos \phi_x}, & \text{where} \\ a_x &= \frac{a_t}{\cos \phi_x} + g \sin \varphi_x, \\ a_y &= \frac{a_n}{\cos \phi_y} + g \sin \varphi_y, \\ \omega_z &= \frac{\omega}{\cos \xi}, \\ \kappa &= \frac{v_x}{\omega_z}, \\ \delta &= \arctan(L_w \cdot \kappa),\end{aligned}$$

$$\begin{aligned}v &= \sqrt{\dot{x}^2 + \dot{y}^2}, \\ a_t &= \ddot{x} \cos \theta + \ddot{y} \sin \theta, \\ a_n &= -\ddot{x} \sin \theta + \ddot{y} \cos \theta, \\ \omega &= \dot{\theta}, \\ \cos \phi_x &= \mathbf{x}_b^T \mathbf{x}_{yaw}, & \cos \phi_y &= \mathbf{y}_b^T \mathbf{y}_{yaw}, \\ \sin \varphi_x &= \mathbf{x}_b^T \mathbf{b}_3, & \sin \varphi_y &= \mathbf{y}_b^T \mathbf{b}_3, \\ \cos \xi &= \mathbf{z}_b^T \mathbf{b}_3,\end{aligned}$$

Control inputs and dynamical variables

Off-Road Navigation Maps for Robotic Platforms using Convolutional Neural Networks

Satellite DL + Low Aerial SfM to Construct Nav Map

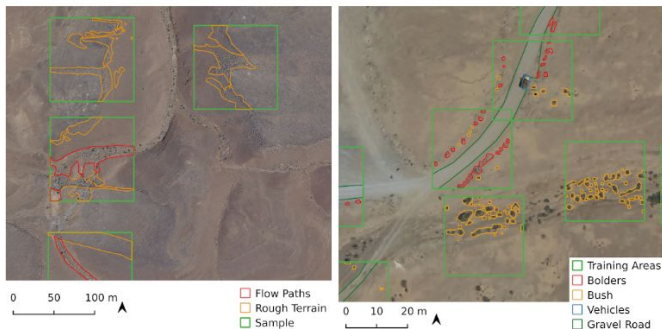


Figure 2. Samples for the training of the macro-feature model (left), the green rectangles denote the extent of the sample while red areas show flow paths and orange areas rough/rocky terrain. The samples for the micro-feature model (right), this includes boulders (red) as well as bushes (orange), gravel roads (dark green) and vehicles (blue). Vehicles were detected to later remove them from roads.

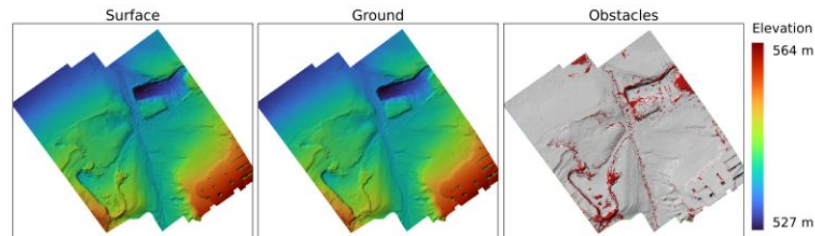


Figure 8. Three different maps created with structure from motion - the digital surface model (denoted as Surface), the digital terrain model (denoted as Ground), and the obstacles detected by subtracting the elevation model from the surface model.

Off-Road Navigation Maps for Robotic Platforms using Convolutional Neural Networks

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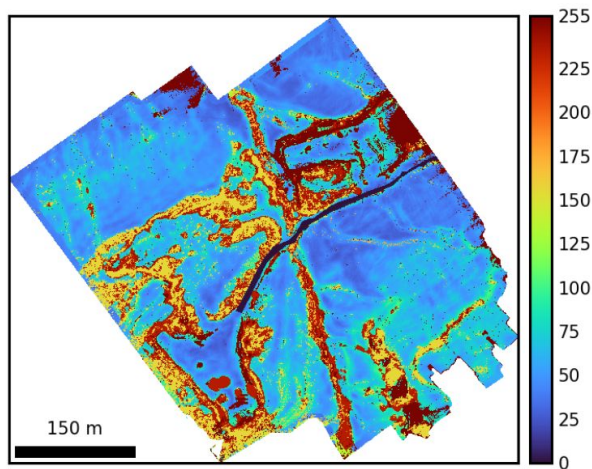


Figure 10. Navigational Map for the base station perimeter. The map is a fusion of the navigation parameters detected via GeoAI and structure from motion



AMADEE-20 Mars Sim Mission Field Test



(11/20)

Weekly Updates

1 COMPA Traversability Optimizer with Holonomic constraints

1.1 Hybrid D* Lite (Global Traversability Estimate)

What we already have (slope + roughness with weights) + maybe some extra stuff in terms of math + adapted to a D* Lite (if possible hybrid to make it a global kynodynamics planner)

1.2 Local Traversability Costs (Primitives?) with NMPC

1.2.1 NMPC Cost:

$$J = \sum_{k=0}^{N-1} \left(\underbrace{\|x_k - x_k^{\text{ref}}\|_Q}_{\text{hybrid D* Lite}} + \underbrace{\|u_k\|_R}_{\text{control effort}} + \underbrace{c_{\text{terrain}}(x_k)}_{\text{accumulated traversability cost}} \right) + \underbrace{\|x_N - x_N^{\text{ref}}\|_P}_{\text{terminal tracking}}$$

$$x = [x_{\text{body}}, y_{\text{body}}, \theta_{\text{yaw}}, v, \omega, \phi_{\text{roll}}, \theta_{\text{pitch}}, \dots]^T \quad \text{Potentially orientation gradients}$$

$$x_{k+1} = f_d(x_k, u_k) \approx x_k + T_s f(x_k, u_k)$$

$$\dot{x} = f(x, u)$$

$$\|u_k\|_R + \Delta u_k^T S \Delta u_k$$

1.2.2 Constraints:

$$|\phi_{\text{roll}}| \leq \phi_{\text{max}}, \quad |\theta_{\text{pitch}}| \leq \theta_{\text{max}} \quad + \text{Velocity constraints?}$$

A minimum distance to untraversable terrain:

$$d(x_k, \mathcal{O}) \geq d_{\text{min}}$$

Actuator limits:

$$\omega_{\text{min}} \leq \omega \leq \omega_{\text{max}}$$

1.2.3 Accumulated Local Traversability Cost:

GMU Traversability cost - could be good when adapted to rocker mechanism (plus needs some sort of accumulation - integrate or sum every visited cell in a given path)

$$c_{\text{terrain}_1}(p) = \mathbf{w}_1 \cdot [\text{roll}, \text{pitch}]^T + \mathbf{w}_2 \cdot [\mu^{\Delta v}, \sigma^{\Delta v}]^T + \mathbf{w}_3 \cdot [\mu^{\Delta q}, \sigma^{\Delta q}]^T$$

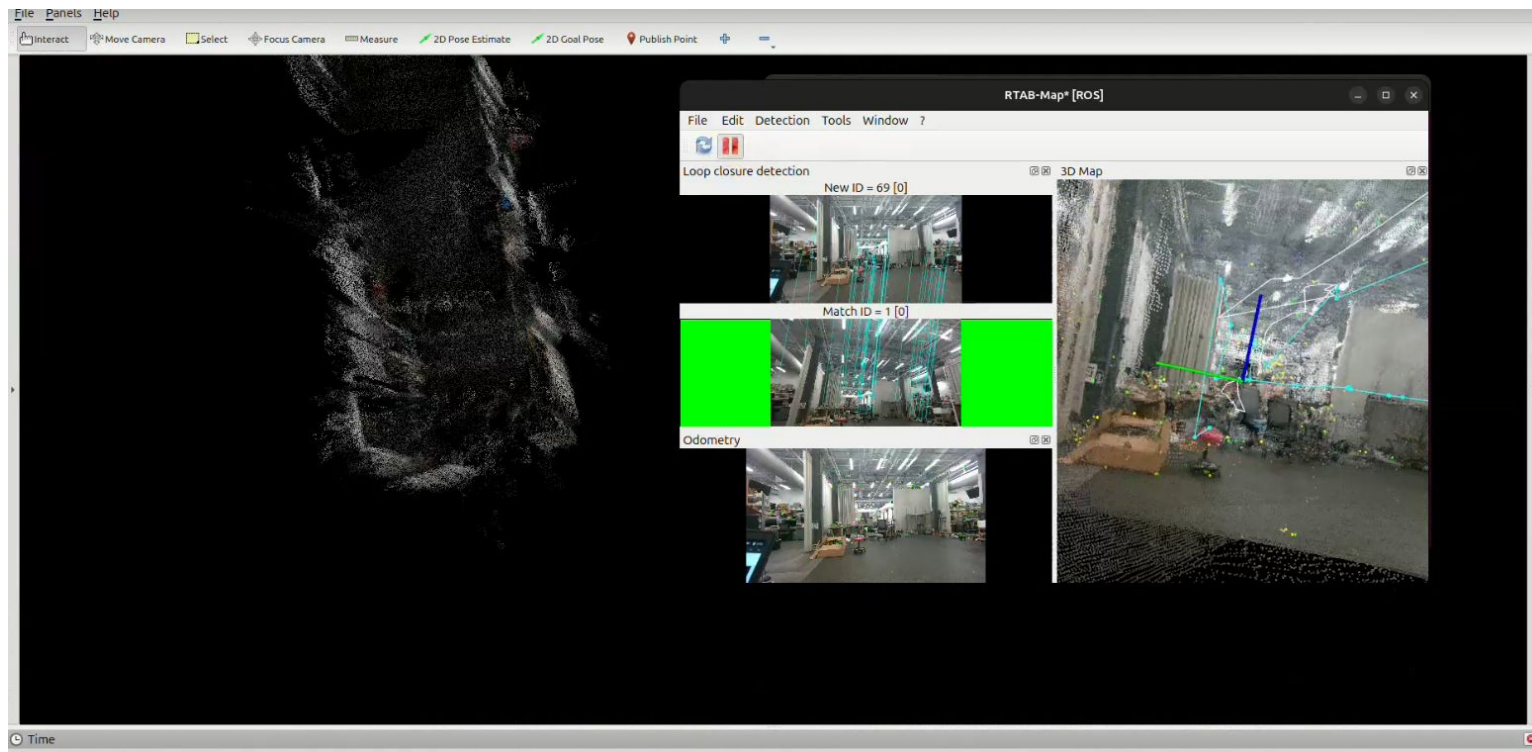
Another interesting cost:

$$c_{\text{terrain}_2}(p) = w_d d + w_s f_{\text{slope}} + w_g f_{\text{gradient}} + w_{\text{slip}} f_{\text{traction}} + w_c f_{\text{curvature}} + w_a f_{\text{attitude}}$$

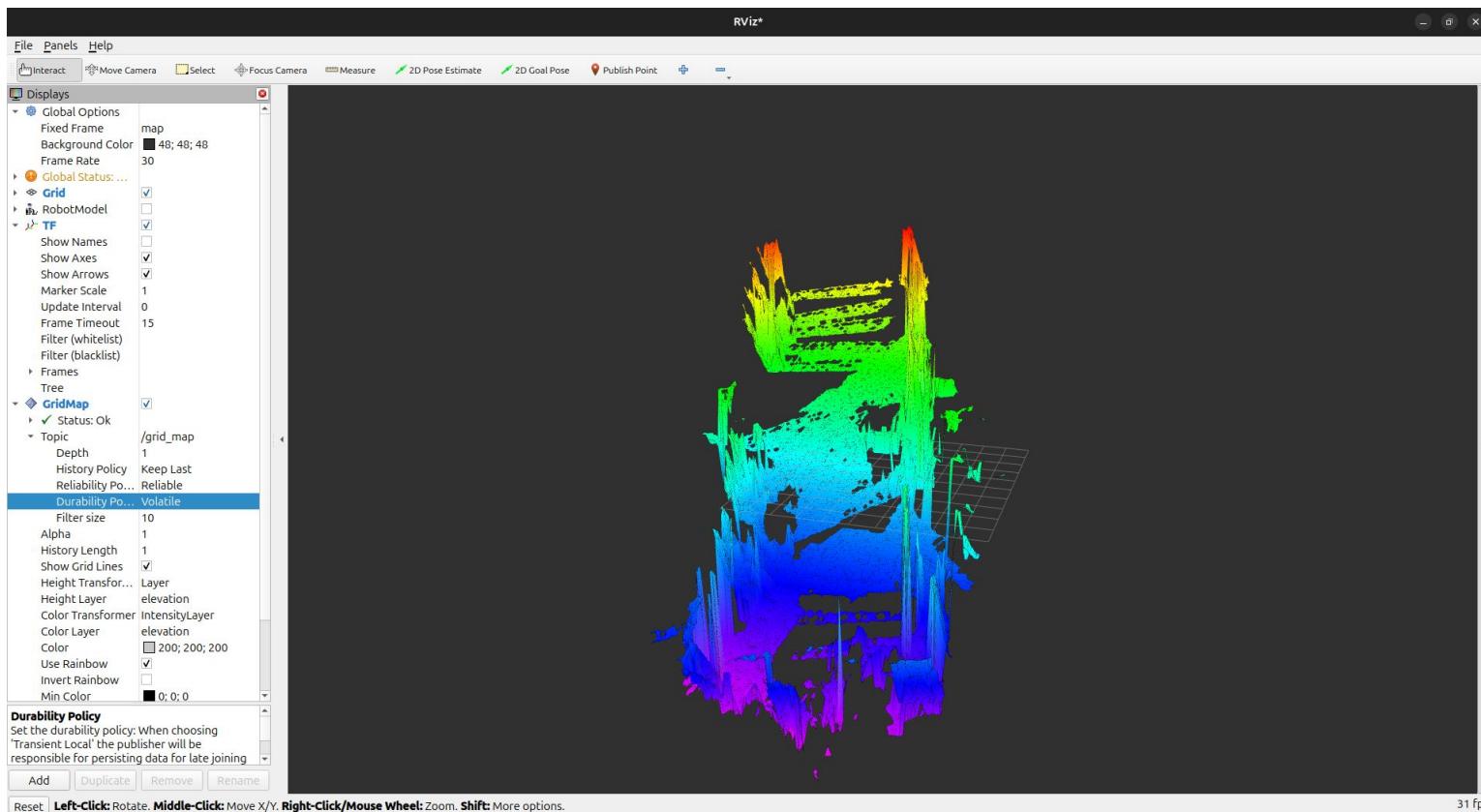
1.2.4 Notes:

Separate the base from the turret+gimbal?

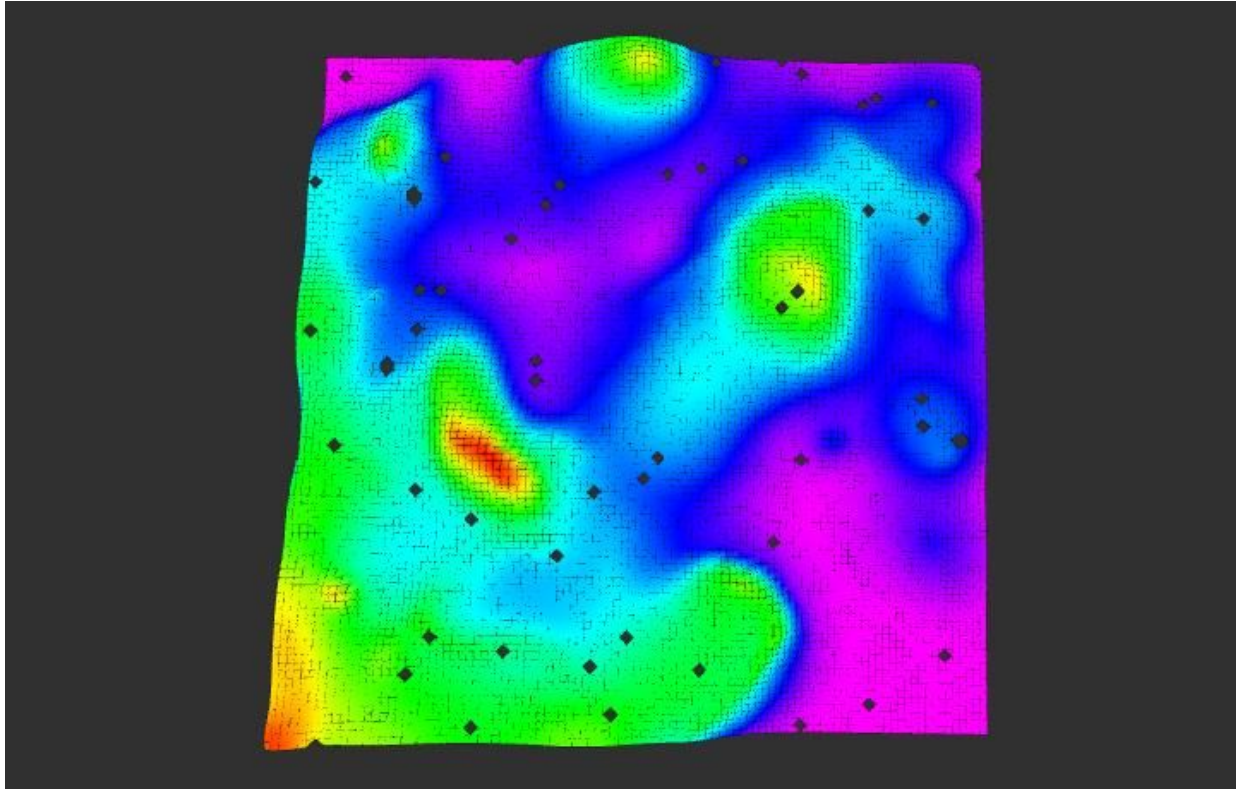
CV: SfM into Grid Map



CV: SfM into Grid Map



CV: SfM into Grid Map



- Use a LiDAR or TOF Camera with [DLIO](#) instead of realsense. Issue because realsense isn't a real kinetic sensor
- Aniket meet
- PCL->gridmap->COMPA_traversable_map
- Caster quaitative
- Coupling between Base and turret/gimbal or simple controls